WORKING PAPERS IN
ECONOMICS AND STATISTICS

Report No 5 | 2026

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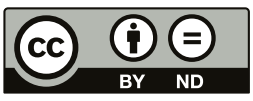
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Series editor: Thomas Giebe

ISBN: 978-91-8082-510-8 (pdf)
DOI: https://doi.org/10.15626/ns.wp.2026.5

Report No 5, Department of Economics and Statistics, Linnaeus University, Växjö, 2026

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Misogyny and xenophobia online: a matter of anonymity[1]

Emma von Essen[2], Joakim Jansson[3][4]

September 21, 2026

[1]We thank Sara Ekman, Therése Jonsson, Didrik Prohorenko, Karin Östling Svensson and Mikael Arvidsson Martins for their excellent RA work. Furthermore, we thank Lina Eklund; Björn Tyrefors; Jonas Vlachos; David Strömberg; Lena Hensvik; Mats Wirén; Jussi Karlgren; David Yang; Dan-Olof Rooth; Björn Öckert; Abdulaziz Abrar Rashid; Alice Wu; Maria Petrova; and Patrick Francois. The financial support from the Swedish Research Council (project number 2013-01789) is also gratefully acknowledged.

[2]Department of Sociology, Uppsala University, 751 26 Uppsala, Sweden, and Swedish Institute for Social Research, Stockholm University, 106 91 Stockholm, Sweden. E-mail: emma.von.essen@soc.uu.se

[3]Department of Economics and Statistics, Linnaeus University, 351 95 Växjö, Sweden, and the Research Institute of Industrial Economics, 10215 Stockholm, Sweden.

[4]Corresponding author: Emma von Essen, Department of Sociology, Box 624, SE-75126 Uppsala, Sweden (email, emma.von.essen@uu.se)

Abstract

Social media rapidly disseminates political opinions and information, serving as a cornerstone of democratic participation. In these online spaces, anonymity can protect free expression, but it may also facilitate hate speech by lowering the social and reputational costs of norm violations. Our study examines how reducing anonymity may shape different forms of prejudiced expression in unequal ways. To explore how anonymity shapes xenophobia and misogyny in online political discussions, we combine quantitative text analysis using fine-tuned BERT models with a difference-in-differences design. We show that reduced anonymity decreases xenophobia but has no comparable effect on misogyny. Moreover, the responsiveness to xenophobic hate speech after a reduction in anonymity seems to stem from individuals with a history of extensive xenophobic writing. Through our exploration, we argue that anonymity's influence depends on the social and reputational costs associated with the various forms of prejudice. The study also demonstrates the value of combining computational text analysis with quasi-experimental methods to study heterogeneous forms of online hate.

Keywords: online, xenophobia, misogyny, anonymity, machine learning
JEL: C55, D00, D80, D90

1 Introduction

Anonymity is common in online spaces where information spreads quickly, such as social media, discussion forums, and website comment fields (Siegel 2020; Jackson Bailey and Welles 2020). It has always been a contentious feature of democracies, known for both protecting individuals' ability to express opinions and engage in discussions without fear of sanctions from governments or society, and yet also for concealing the user's personal identity (Marx 1999; Boyd and Field 2016). Concealing identity reduces accountability and can lead to prejudice-based hate speech (Citron 2014; Levmore and Nussbaum 2012), and much research confirms that high levels of anti-normative behavior occur in more anonymous settings (Moore et al. 2012; Van Royen et al. 2017; Hirsh Galinsky and Zhong 2011). Moreover, hate speech begets hate speech, generating a large volume of hate online (Cheng et al. 2017; Álvarez-Benjumea 2023).

A large amount of hate speech can silence affected groups by driving them to withdraw from civic and political participation (Barnidge et al. 2019; Henson Reyns and Fisher 2013; Pew Research Center 2014; Pew Research Center 2021; Pew Research Center 2022). Such withdrawal implies less diversity in civic discussions, which can lead to lower political accountability or less informed decisions (Strömberg 2015). Which groups are silenced will likely depend on the suppression of different social norms against prejudice, which signal possible social repercussions for expressing such views (Álvarez-Benjumea 2025).

While anonymity is known to facilitate hate speech, it remains unclear whether reducing anonymity restrains all forms of prejudice-based hate speech equally (Siegel 2020). For example, misogyny in online discussions seems to be socially normalized (Citron 2014). If such normalization differs from that of xenophobia, it suggests that the responsiveness to greater reputational risk will be higher among those expressing xenophobic views than among those expressing misogynistic hate. To address this question, our study examines how anonymity affects online hate speech across stereotypes, using a sudden breach of anonymity on Flashback, Sweden's largest, broadest, and oldest civic discussion forum.

To guide our expectations, we use social identity research linking online hate speech behavior to responsiveness to reputational concerns. Disinhibition and social identity theories suggest that anonymity facilitates expressions that violate broader social norms by weakening social sanctions from the surrounding society and shifting users' attention to the forum group (Suler 2004; Postmes Spears and Lea 1998; Reicher Spears and Postmes

1995). A person's sense of self and self-esteem are assumed to lie in the chosen group, so they adopt the group's values and norms.

Forum-specific norms quickly replace societal norms related to hate speech (Álvarez-Benjumea 2023). Moreover, when the perceived societal repercussions of hate speech are muted, what others say and do at the forum, i.e., the specific forum-generated norms, are found to drive hate speech provision rather than forum sanctions (Álvarez-Benjumea and Winter 2018). However, if anonymity reduces fear of reactions from social peers outside online spaces associated with expressing prejudice (Álvarez-Benjumea 2025; Munger 2017; Matias 2019; Hirsh Galinsky and Zhong 2011), that fear likely depends on how socially accepted the prejudice is in the surrounding society.

Based on social identity theories, Ali and Bénabou (2020) developed a principal-agent model that formally links anonymity to social image and reputational concerns. Applying their framework to our setting, governments or online platforms (the principal in their model) can reduce anonymity, influencing individuals' (the agents in their model) propensity to express prejudiced views. Agents may reduce their prejudiced expression when anonymity decreases because of the risk of reputational damage. However, if the variation in how agents perceive the social or reputational costs of expressing a specific prejudice is greater than that for another prejudice, reducing expressions of that specific prejudice will send a less clear signal to others that an agent is making an "honest" change in prejudice. This suggests that because others will find the users' changed behavior harder to evaluate as a true change in beliefs, reducing prejudiced expressions will yield fewer reputational gains for the agent. The model claims that this, in turn, will lead to less responsiveness to reduced anonymity on average. Following Ali and Bénabou (2020), if misogyny is more normalized in online settings (Citron 2014:p. 18) than is xenophobia, the influence of a sudden drop in anonymity is expected to be larger for xenophobic hate speech than for misogynistic hate speech.

To evaluate this expectation, we utilize a sudden decrease in anonymity on the discussion forum Flashback. The event involved journalists increasing the risk of public exposure of a specific group of users' prejudice, which substantially increased those users' reputational risk. This provides an opportunity to examine how heightened reputational risk shapes online expression using a natural experiment. To use the event as such, we generated machine-learning measures of xenophobia and misogyny (using BERT models)

as inputs to canonical difference-in-differences regression models.[1] Through these models, we test the expectation of differential influence of anonymity on xenophobia and misogyny. Moreover, individuals with more hate speech written before the event should face a larger probability of being exposed. We thus also explore how the extent of an individual's written hate speech before the event contributes to changes in expressing prejudice. Lastly, if individuals use anonymity to express general hate or frustration, and expressions of xenophobia and misogyny are seen as interchangeable, reduced anonymity may give rise to individuals shifting their type of hate speech. We also explore whether reduced anonymity gives rise to individuals shifting their expression from one prejudice to another following the event.

We show that the behavioral effects of reduced anonymity depend on the type of prejudice and individual past hateful behavior. Reducing anonymity lowers online xenophobia but not misogyny. We conclude that the expected social reputational costs of expressing xenophobia and misogyny differ, since it was those individuals who had written an extensive amount of xenophobic posts before the event who decreased their writing following reduced anonymity, not those with fewer prejudiced writings.

The study makes two key contributions. First, the study offers new insights into how online predjudice-based hate speech interacts with societal norms. Previous studies have explored anonymity related to specific prejudices (Álvarez-Benjumea 2023) or compared prejudices online unrelated to anonymity (Siegel 2020; Saha Chandrasekharan and De Choudhury 2019). Complementing these literatures, we offer a deeper understanding of how anonymity affects individuals' willingness to express online hate by showing how the consequences of reduced anonymity vary across forms of hate speech. Distinguishing among different forms of prejudice is therefore important for understanding when and why anonymity affects online hate. Second, the study contributes methodologically to research on xenophobia and misogyny in online discussions. Despite the abundance of data available online and the progress of quantitative analytical tools (e.g. Keuschnigg Lovsjö and Hedström 2018; Molina and Garip 2019), much research tries to unpack the problem of misogynistic and xenophobic hate speech in discussions online using qualitative approaches (Bliuc et al. 2018; Daniels 2013), and critical discussions on the study of xenophobia and prejudice have called for the development of quantitative approaches in sociology (Quillian 2006). Our study responds to this call by quantitatively highlighting

the complex differences behind xenophobia and misogyny online.

Throughout the paper, we employ the terms misogyny and xenophobia for conciseness. Consistent with previous research on online hate speech (Álvarez-Benjumea and Winter 2018; Álvarez-Benjumea 2023), we use hateful language and hate speech interchangeably to refer to hostile and toxic language directed at individuals within a social group. The concepts draw on Allport (1954) (p.7)'s classical definition of prejudice as antipathy based on faulty or inflexible generalizations. Prejudice therefore requires both a target group and an associated belief about that group's behavior or attributes. In this study, we combine prejudice with antipathy, implying that hate speech may not involve an articulated stereotype of the target group.

2 Background

In this background, we first briefly lay out the social science arguments for why we should care about online hate and why users generate such hate speech. Then we apply our setting to a formal model to generate empirical predictions. Lastly, we argue that the Swedish forum offers a relevant case for studying anonymity and online hate.

Online hate is a contentious issue because it not only challenges democratic values but also harms victims' health and labor market opportunities (Siegel 2020; Wu 2020; Saha Chandrasekharan and De Choudhury 2019). As has been shown for traditional media, hate speech can trigger acts of violence and hate (Yanagizawa-Drott 2014; Adena et al. 2015; DellaVigna et al. 2014); what is written and shared online seems to translate into political and social actions outside the Internet (Zhuravskaya Petrova and Enikolopov 2020). These effects include voting, protests (Acemoglu Hassan and Tahoun 2018; Qin Strömberg and Wu 2017; Enikolopov Makarin and Petrova 2020), attitudes toward government (Enikolopov Petrova and Sonin 2018), and racial hate crimes (Chan Ghose and Seamans 2016; Müller and Schwarz 2021; Müller and Schwarz 2020).

The vast amount of hateful language online reinforces already existing stereotypes, which in turn can produce a normalization of hateful speech, not only online but in society at large (Citron 2014; Fox Cruz and Lee 2015; Álvarez-Benjumea 2023). Normalization processes of dominant social stereotypes such as those based on race and gender (Siegel 2020; Silva et al. 2016; Allcott and Gentzkow 2017) may differ across stereotypes and

contexts. To understand these processes, we need to know more about who generates hate and why.

Because of user anonymity, we know more about who is targeted by online hate than about those who generate it (Siegel 2020; Citron 2014). In general, two main standpoints exist regarding online haters: anyone can become a hater, or people are born haters. Cheng et al. (2017) argues that anyone can start hating online because the context fosters hate. Others argue that there are common denominators among people who hate online, suggesting, for example, that men are more prone to generating as well as finding hate acceptable than women (Costello and Hawdon 2018; Costello et al. 2019).

One of the most common explanations for hate speech in anonymous online settings is online disinhibition (Suler 2004), in which the absence of social consequences makes some people more likely to write hatefully and aggressively. What users write under an anonymous handle does not match what they would write publicly under their legal name (Moore et al. 2012; Van Royen et al. 2017). (bursztyn2020extreme) supports this, finding that expressing extreme opinions is closely tied to expectations of social and governmental repercussions for deviating from the norm.

Moreover, online anonymity is rarely binary, in the sense that a user's name is either exposed or not (as the sole marker of identity). It is rather contextual (Eklund et al. 2022) and depends on whether others can identify various aspects of who we are and how such identification can occur. Many of the multi-layered facets of anonymity are features controlled by online platforms. This implies that these features may be altered by internal or external policy.

However, if a government implemented a policy that reduces online anonymity, it is unclear whether the effect on hate speech reflects changes in users' prejudice or simply a suppressing effect that changes their writing behavior (Flores 2017). Moreover, the fear of repercussions for writing hateful speech may also lead individuals in online communities to suppress less socially acceptable opinions (Jackson Bailey and Welles 2020; Zhang and Kizilcec 2014).

2.1 Conceptual framework

Our setting examines anonymity as a facilitator of online hate that may vary by the prejudice expressed. The underlying assumption is that a change in anonymity creates

a risk of social exposure that, in turn, can lead to sanctions for transgressing the norm against writing xenophobic and misogynistic content.

To formalize our expectations, we use the principal-agent theory of Ali and Bénabou (2020) as our main conceptual framework, which clearly connects social norms to reputational concerns. Agents' concerns about social image are central to implementing social behavioral changes. In this framework, agents can contribute to a public good that generates positive externalities. Contributing to the public good is costly, but it can also improve social reputation. The principal can affect contributions to the public good by increasing the visibility of who contributes or not. In our setting, the public good with positive externalities is that users refrain from misogynistic or xenophobic writing. Online discussants are agents who contribute by writing messages. Their choices depend on their views on prosociality, writing non-hateful messages, and reputational concerns. The principal can be a government, authority, or platform aiming to reduce online xenophobia and misogyny. Anonymity may influence how agents evaluate contributions, as people care about others' perceptions and want to appear prosocial.

The theory predicts that increased public good provision (less hate speech) occurs as anonymity decreases, but differences in social image concerns affect this pattern. High dispersion in reputational concerns and perceptions of social punishment makes it hard for others to distinguish true prosociality from reputational concern. Greater dispersion will generate less responsiveness to anonymity. If xenophobia and misogyny differ in such dispersion because, for example, exposure risks and social costs differ, it leads to differentiated changes in public good provision. A differentiated response to anonymity between xenophobia and misogyny would thus suggest that writers either differ in their reputational concerns about each form of hate speech or that there are different expectations about the repercussions of making such hate speech.

The model doesn't specify what could drive such differences in dispersion. Reputational concerns emerge from the social context of online discussion environments, alongside the surrounding society. General discussion forums like Reddit and Flashback constitute a broad type of online community. They attract a large number of users who, under anonymity, want to discuss different topics for various reasons.

2.2 The social context: A large and broad discussion forum

We argue that Flashback provides a suitable context for exploring how anonymity links to various forms of hate speech.

First, Flashback is a well-known, widely used online discussion forum established around 2000. A survey by Davidsson Palm and Melin Mandre (2018) found that 33% of the Swedish population reported having actively used Flashback, with a higher share among men than women (40% vs 26%). Moreover, users expect to be anonymous, since Flashback is known for upholding user anonymity and promoting freedom of speech. Also, at Flashback, all writings are publicly available to read, but a person needs an account to write. Creating an account costs nothing; users need only an e-mail address and a self-created, anonymous username. All discussions are publicly available and are never deleted, even after the user terminates their account. Many individuals and organizations never create accounts and only read the discussions.

Second, the structure resembles a typical broad discussion forum (message board). It resembles Reddit, which was established a few years later.[2] These forums host a wide range of comments, hateful and non-hateful.[3]

The messages are presented chronologically and the posts are organized into subforums covering topics ranging from politics, sexual preferences, and drug abuse to electronics and family relationships. Each subforum is divided into threads, each a separate discussion of a specific topic (or question). Users can contribute to a thread by writing a new post or replying to a previous one. Discussions occur within threads, not across them. Posts/messages are displayed in chronological order, along with the time of entry, the username, the number of posts by the user (at the time of posting), the user registration date, and sometimes a self-selected picture. The first twelve posts are automatically shown on the screen when clicking on a thread.

Third, there is no moderation of user-generated content. Each subforum has users with a moderator role who oversee discussions in accordance with internal rules (netiquette). If a user violates any rules, the moderator can issue a warning or temporarily or permanently ban the user from the forum. The general rule at Flashback is that all opinions are welcome. Most warnings and bans are issued primarily for content labeled "off-topic" when a discussion strays from the thread's original focus. To understand the structure of the discussion forum, we interviewed one Flashback moderator to learn about

the moderators' influence on how debates are shaped. The interviewee mentioned that moderators primarily help maintain users' sense of anonymity.

The moderator's claim is supported by the rarity of users receiving warnings or being banned for generating hateful content, such as misogyny and xenophobia. During our data collection, a simple forum search showed that 623 users were banned for advertisements and 280 for copyright infringement. This is a stark contrast to the 36 cases of users being banned for writing hateful posts against minorities. Another common violation is the misuse of multiple accounts, with 421 users banned for this reason. Moreover, during our review period, Flashback did not change its internal rules. Note that in a Swedish context, hate speech can be illegal, mainly in relation to crimes related to discrimination. However, hateful speech, such as expressing xenophobic and misogynistic views, is not illegal.

Fourth, using data from a traditional discussion forum offers additional advantages over other social media platforms. One advantage is that its user base is likely broader than on other discussion boards, such as incel communities where misogyny is common (Halpin et al. 2025). Moreover, there is no algorithmic prioritization, so content appears in chronological order. Therefore, content and individual user behavior are not shaped by a background algorithm, unlike data from the platform X. Also, users cannot delete their old posts, so they cannot later present themselves as less hateful writers. Finally, because of the forum's minimal moderation policy, we can analyze posts as users originally wrote them.

3 The event: The unexpected decrease in anonymity

On September 10, 2014, journalists unexpectedly revealed that they had obtained the identities of at least one-third of users who registered on the forum before March 2007. Individuals could easily determine if they were at risk of exposure from a list of usernames. The owner of Flashback also publicly confirmed that users registered before March 2007 were at risk of having their writings and identities exposed. In section S.6 of the Supplementary Material, we detail how the journalists technically could link users' identities to their usernames and how forum owners could verify who was at risk.

The event posed a credible threat to the users on the list. The journalists, named

"Researchgruppen", were previously known for exposing users who made xenophobic or misogynistic statements. From Flashback, they also published articles revealing the identities of four individuals and their controversial writings, leading to the loss of two politicians' jobs. The journalists' main goal was to reduce hateful and disrespectful online language, noting that none of the exposed users posted potentially illegal content.

Moreover, to illustrate that the threat was unexpected and how widely it was noticed, Figure 1a shows Google Trends data on weekly search frequencies for Flashback and Researchgruppen from 2013 to 2017. Two peaks emerge: the first in September 2014, during the media revelation, and the second in February 2015, when some users were identified by name and hate content was reported in the media. Figure 1b instead shows the monthly number of news articles in Sweden containing these words, again with peaks in September 2014 and February 2015.

In the paper, we treat the first peak as the event's start. Thus, from September 10, 2014, users registered before March 2007 faced a higher risk of public exposure of their identities and of Flashback content. Those registered after March 2007 did not.

[Insert Figure 1 here]

In the study, we treat all estimates as intention-to-treat (ITT) estimates. This means we assume that users before March 2007 were exposed to risk (treated) and that other users were not (untreated). In the end, this type of analysis may slightly underestimate the potential effect sizes.

To assess whether users at risk understood the threat as critical, we manually examined the first four pages of posts in the thread discussing the revelation that journalists had obtained the identities of Flashback users.[4] The thread began with the administrator assuring users that Flashback had increased security to minimize the risk of future similar events and, thus, of other users being exposed. The administrators took no measures to reduce harm to users registered before March 2007.

The discussion that followed in the thread called for measures such as censoring usernames or deleting users' posts. The main argument from users concerned the fear of social consequences if their names and writings were exposed. The following entry in the thread, by a user registered before March 2007, exemplified the concern among users at

risk of exposure.[5] "[...] Forever being a pariah when Googled is bad enough, but to also risk being severely beaten or murdered if you are exposed is so serious that the negative consequences are impossible to grasp at present." We thus argue that, given the extensive discussions both on Flashback and in traditional and social media, no user should have missed whether they were at risk of exposure.

It is also theoretically possible that users in the not-at-risk group also believed that their risk of exposure increased after the event. If so, and if these users then changed how they wrote on Flashback, our analysis might underestimate the behavioral change we aim to capture. However, there was only a small decrease in the number of xenophobic, misogynistic, and total posts among those not at risk, suggesting that perceived risk was a minor or nonexistent issue (see Figure 3 and the Supplementary Material Table S.4).

4 Study design

This study uses machine learning and regression models to analyze how anonymity affects xenophobic and misogynistic language in online forum messages from 2012 to 2016. It first quantifies messages with machine learning and extracts variables, then applies difference-in-differences regression models.

We obtained the data from three subforums with the most extensive political discussions: domestic politics, immigration, and gender and feminism. An example of a thread in the domestic politics forum is "Keywords for a political alliance", in which participants discussed whether all Swedish politicians have similar attitudes toward gender quotas and immigration. In another thread titled "The number of sick days has dropped more than 50%", the discussants focused on paid sick leave and people being forced to work while ill. In the feminism subforum, threads included discussions of how feminists affect Swedish politics, such as "Do feminists want gender equality?" (authors' translations of the thread labels).

4.1 Generating quantitative measures from text

For each post, we both extracted metadata, including the subforum name, thread title, username, post timestamp, the user's registration date (start date), and the full message text. Quantifying the message text into variables of xenophobia and misogyny involves

three steps: manually classifying a subsample, training a classification model on that sample, and running the model on the full text data.

Step 1: The manually classified dataset comprised 4,040 posts, evenly distributed across the three subforums. To generate these classification measures, a research assistant manually classified the first twelve and last five posts from 100 randomly selected threads within each subforum.[6] The research assistant evaluated each post holistically, classifying it as hateful or non-hateful and identifying the target group of hate-related content. This classification process followed detailed guidelines derived from prior research on automated hate speech detection (e.g., Cohen et al. 2014), see Supplementary Material, Sections S.1 and S.7. Posts expressing hatred or animosity toward women were classified as misogynistic, whereas those targeting foreigners and ethnic minorities were categorized as xenophobic. The classification instructions also included criteria for identifying threatening and aggressive content.

To illustrate the nature of posts classified as xenophobic and misogynistic, we present two author-translated examples. It is important to note that posts classified as misogynistic or xenophobic often contain offensive and disturbing content. One post categorized as misogynistic stated "[...] Women are biologically built to obey and receive orders [...]." Similarly, a post classified as xenophobic reads "[...] The wog should preferably be dead, but it is not necessary. The important thing is that the wog ends up six feet under."

To mitigate potential bias from human annotators (Sap et al. 2021), we used standardized classification guidelines, as noted above, and conducted an inter-rater reliability assessment in which another independent evaluator classified posts. The agreement rate between the two annotators was 94% for xenophobia and 95% for misogyny. The annotators' lack of awareness of the research question also mitigated bias. Table S.5 in the Supplementary Material presents the descriptive statistics for the classified data.

Step 2: To transform the remaining textual data into quantifiable measures, we used the annotated dataset to fine-tune a pre-trained machine learning model, specifically the Bidirectional Encoder Representations from Transformers (BERT) model, to predict instances of xenophobia and misogyny. BERT is a large language model designed for fine-tuning on domain-specific text classification tasks. Do Ollion and Shen (2024) demonstrates that BERT produces predictive performance comparable to or exceeding that of human annotators.[7] While a key limitation of BERT is opacity, i.e., the challenge

of determining the precise reasoning behind its classifications of specific entries as xenophobic or misogynistic, BERT models typically provide high predictive accuracy, meaning that they generate a high share of correct classifications.

Any approach to measuring text content will inherently involve some degree of error. The critical question is how to minimize the impact of such errors. To this end, we chose to optimize the xenophobia and misogyny models by calibrating them to maximize the sum of the true positive rate (sensitivity) and the true negative rate (specificity). Further details on this process are provided in the Supplementary Material, where we also discuss measurement error issues in depth and explain why these are unlikely to affect our results in a qualitative way (see sections S.2, S.2.1 and S.3).

Step 3: We then applied the calibrated prediction models to the full dataset and classified all posts recorded in our data, covering the period from the beginning of 2012 through the end of 2016. This approach substantially expanded our sample size and significantly enhanced our capacity to analyze the core research question. The Supplementary Material details the procedure used to assess the quality of the model-generated predictions (see section S.2.1).

There are a number of concerns when quantifying measures using text. The first concern is that users at risk of exposure could change their written language to express xenophobia or misogyny more frequently than users not at risk following the event. This is not a major issue in our setting, as we calibrated the BERT model using data coded by a research assistant (RA) who was unaware of the specific research design and whether users were at risk. Second, other statistical learning models may not generate the same measures, but when we use a simpler LASSO model to predict xenophobia and misogyny, we obtain qualitatively similar results (Supplementary Material, Tables S.9 and S.10). Finally, most classifications can be subject to measurement error. In our case, we also use the total number of entries as an outcome variable, which does not have such errors.

4.2 Sample construction and description

We constructed two distinct data panels to examine how unexpected changes in user anonymity influence users' xenophobic and misogynistic writing on Flashback. The first dataset is a balanced two-period panel dataset with the individual user as the unit of analysis. To build the balanced panel, we restricted the sample to users who posted at

least once in both the pre- and post-event periods. We then collapsed the data to produce a single pre-event and a single post-event observation per user. The second dataset is an unbalanced panel comprising all posts written during the study period.

[Insert Table 1 here]

Table 1 shows summary statistics for both the balanced (Panel A) and unbalanced (Panel B) panels.

The balanced panel consists of 11,159 unique users, yielding 22,318 observations across the two periods. The outcome variables include both the number of posts containing xenophobia or misogyny, as well as the proportion of each user's posts classified as containing xenophobia or misogyny. We also explore the total number of posts authored per user in each period as an outcome variable. All outcome measures are aggregated at the user level within each period and reflect individual patterns of writing messages. On average, this subset of users, those active both before and after the anonymity-related event, authored posts containing xenophobia in 29% of cases and misogyny in 21% of cases.

The unbalanced panel includes all posts during the study period: 1,984,224 posts authored by 48,672 users. Posts constitute the unit of analysis and capture the prevalence of xenophobic and misogynistic content encountered by readers on Flashback. In this dataset, the outcome variables include both the number of posts containing xenophobia or misogyny, as well as the proportion of such posts relative to the total number written. Many users engage with the platform solely as readers without contributing to discussions. Among these, 33% were classified as containing xenophobia and 26% as containing misogyny. Notably, these proportions are broadly consistent with those observed in the balanced user panel.

To understand the amount of xenophobia and misogyny these forums can produce and thus, how much common readers of Flashback face, we used traffic statistics from Alexa.[8] A typical user views approximately seven pages per day, with each page displaying 12 posts. This browsing pattern suggests that the average reader is exposed to 84 posts daily, of which roughly 27.7 contain xenophobia, and 21.8 contain misogyny.

4.2.1 The distribution of posts and hate

Before we answer the main research question linking misogyny and xenophobia to anonymity, this section will show that posting in general, and hateful posting in particular. Moreover, we show that producers of xenophobic and misogynistic content do not seem to overlap.

Most users produced few or no posts containing xenophobic or misogynistic content, suggesting that a minority of highly active users wrote most of these posts. Figure 2 shows the distribution of five user-level measures: total posts, posts containing misogyny, posts containing xenophobia, and the corresponding shares of misogynistic and xenophobic posts.

[Insert Figure 2 here]

In order to explore the general relationship between misogyny and xenophobia as encountered by forum readers, we examined the raw correlation between the two at the post level. The analysis revealed a small positive relationship, with a coefficient of 0.03. This suggests that while there is a slight tendency for posts containing xenophobia to also include misogynistic content, such co-occurrence is relatively uncommon. See Table S.14 in the Supplementary Material.

Interestingly, active writing is not strongly correlated with hate speech. There is a positive correlation between overall posting and the share of posts containing xenophobia or misogyny. However, this relationship is modest. For instance, writing 1,000 additional posts is associated with only a 0.01 (or one percentage point) increase in the share of a user's posts that are classified as xenophobic. The Supplementary Material reports the full results in Table S.14.

5 Analytical approach

To answer our main research question on the differentiated response of xenophobic and misogynistic hate to changes in anonymity, we use two regression-based difference-in-differences methods that are conceptually similar but use different data panels, as outlined below.

Both leverage the same event that happens on a single occasion to one specific group of

users. Our setup is therefore a typical example of the canonical difference-in-differences setting, with one group receiving treatment from a single point forward and a control group that is never treated. Thus, we do not need to employ any new regression techniques for panel data estimation with staggered designs, as our setting does not use staggered events (Chaisemartin and D'Haultfœuille 2023).

We obtained results using the statistical software R. Throughout the paper, we used a significance level of 0.05.

5.1 Method using the balanced panel of individual writers

To estimate the effect of reduced anonymity, we employed a difference-in-differences (DiD) approach, formally specified in Equation (1). The balanced panel captures the effect of reduced anonymity, encompassing changes in individual writing behavior.

$$Y_{itg} = \alpha + \beta Risking\,exposure_g * Post_t + \theta Post_t + \gamma Risking\,exposure_g \qquad (1)$$

$$+ startdate_i + statdate_i^2 + \varepsilon_{itg}$$

Y_{itg} is the outcome variable of individual writer $\imath$ at time $t \in \{pre, post\}$ belonging to group $g \in \{Riskingexposure, notatrisk\}$, i.e. the group of the writer. $Riskingexposure_g$ is a dummy variable with a value of 1 if the user registered before March 2007, $Post_t$ is a dummy variable with a value of 1 if the post was written after the event on September 10, 2014, and $startdate_i$ and $startdate_i^2$ are the user's start date in linear and quadratic form. Finally, ε_{itg} is the error term. In this specification, β captures the effect of the change in anonymity, more specifically, the increased likelihood that a user's identity and associated hateful content could be exposed in traditional media, on subsequent writing behavior. The standard errors are clustered at the individual level to account for dependencies between individual observations.

Because assignment to the treatment condition, being at risk of becoming exposed, was fully determined by users' registration dates, we can, in principle, control for all unobserved differences by including flexible control functions in the user registration date. Accordingly, the main tables present estimates that include linear and quadratic controls for users' registration dates. The results remain consistent across specifications with and without these controls, indicating a strong robustness of our estimates (see Tables S.9, S.11 and S.12 in the Supplementary Material).

5.2 Method using the unbalanced panel of entries

To assess how the event affected the content a typical reader encountered, we turn to the unbalanced panel, which includes all posts within our sample period. The unbalanced panel captures the total effect of reduced anonymity, encompassing changes in both individual writing behavior and user attrition.

As with the balanced panel, we apply a simple difference-in-differences (DiD) estimation strategy to examine how overall discourse on Flashback evolved following the anonymity shock. In this section, however, our primary focus shifts toward the share of content to which readers are exposed, though we also document changes in the volume of posts.

To estimate effects on posting volume, we follow a two-step aggregation procedure similar to Pettersson-Lidbom and Skogman Thoursie (2013). Specifically, we collapse the data to the weekly level by group $g = Risk\,exposure$ and $not\,at\,risk$. This results in two group-level time series measured at weekly intervals. We then reformulate Equation (1) to express the difference in outcomes between the two groups over time:

$$\bar{Y}_{t1} - \bar{Y}_{t0} = \Delta Y_t = \pi + \beta Post_t + \Delta\varepsilon_t. \tag{2}$$

Estimating Equation (2) is analytically equivalent to the individual-level specification in Equation (1), but it is implemented using aggregated weekly data at the group level. In this formulation, the coefficient β retains its interpretation as the difference-in-differences estimator.[9]

6 Results

In this section, we show that reduced anonymity decreases online xenophobia, but not misogyny. We present the results in two parts. First, we examine the effect of reduced anonymity on individual's writing behavior using a balanced panel of individual writers. Second, we examine the effect of reduced anonymity on total writing activity using the unbalanced panel of posts.

6.1 The effect of reduced anonymity on individual writing behavior

In this subsection, we show two things. First, the event decreased the number of xenophobic entries and the total number of entries that users wrote, but did not affect the number of misogynistic posts. Second, we demonstrate that the drop in the number of entries after the event was only present among individuals with high levels of xenophobia before the event.

Table 2 presents the balance test for the pre-event differences in outcomes between users at risk of exposure and those not at risk, controlling for user registration date and its square (*startdate* and $startdate^2$).[10] Across all specifications, we find no statistically significant pre-event differences between the two groups. The key identifying assumption underlying difference-in-differences (DiD) estimation is that, in the absence of the treatment, in this case the exposure of user identities, the treated and comparison groups would have followed parallel trends. Comparable pre-event levels suggest that the groups are balanced on observable characteristics and therefore more likely to follow similar trends over time.[11]

[Insert Table 2 here]

Figure 3 illustrates the average change in the number of posts from the pre- to the post-period per exposed/not exposed users. Panel A shows a clear decrease in xenophobic posts among users at risk of exposure, but no significant change among users not at risk from pre to post. For the number of misogynistic posts, we instead see two significant but small decreases in Panel B. Finally, in Panel C, we again see a clear decrease for the total number of entries, regardless of entry classification, among users at risk of exposure, but no significant decrease among users not at risk.

[Insert Figure 3 here]

To provide a numerical estimate of the effects, the cells in Table 3 present estimates from separate regressions, representing the β-coefficient from Equation (1). Panel A, columns one and two present the effect of the event on the number and the share of xenophobic entries. Panel B presents the same for misogyny, and Panel C displays the

results for the total number of entries regardless of entry classification.

[Insert Table 3 here]

In the first two columns of Panel A, individuals at risk of exposure write fewer xenophobic posts after the event than those not at risk. The effect corresponds to 7.5 fewer xenophobic posts written by users at risk of exposure. However, their share of xenophobic entries remains the same, since users write fewer posts overall. Panel C shows that the total number of entries decreases by almost 15 posts written by users at risk of exposure. For misogyny, we do not see any decline. See Panel B.[12]

Note that the users do not abandon the forum, since by construction all users in the balanced panel wrote at least one post before and one post after the event and thus remain active to some extent.[13]

6.1.1 Who changes their individual writing behavior?

Here, we show that the decrease in the number of posts after the event occurs only among individuals with a high level of xenophobia before the event.

Individuals who wrote xenophobic or misogynistic posts before the event are more likely to perceive themselves as vulnerable to identification by journalists. If these individuals reduced their activity after the event, it would suggest a behavioral response driven by anticipated social repercussions.

To explore whether users with a history of xenophobic or misogynistic posting changed their behavior more than others, we divided the balanced panel based on whether the share of posts containing xenophobia or misogyny in the pre-event period was above or below the median, and then ran separate difference-in-differences models for these two groups (above and below median). This analysis is equivalent to a difference-in-difference-in-difference design, where we assess the interaction between prior engagement in xenophobic or misogynistic content, group assignment (at-risk of exposure vs. not), and the post-event period. It also serves as a robustness check by adding another dimension of comparison.

Why did we divide the sample based on whether xenophobia and misogyny were above or below the median? Going back to Figure 2, we saw that the distribution of xenopho-

bic and misogynistic content is highly skewed: users in the lower half of the distribution contributed few, if any, such posts. Based on this distribution, we categorized users into the top and bottom halves by their pre-event share of xenophobic or misogynistic posts. Table 4 reports results for users in the bottom half of the distribution in columns 1 and 3, and for those in the top half in columns 2 and 4.

[Insert Table 4 here]

Comparing the first and second columns, we see that the decrease in the number of posts following the event occurs only among those with a history of writing many xenophobic posts. In this group, the effect corresponds to a decrease of almost 29 posts, or twice the decrease among all at risk of exposure in Table 3.

Individuals with either high or low pre-event levels of misogyny have similarly sized coefficients. These estimates are also of similar magnitude to the average effect on the number of posts observed in Table 3.[14]

Users who engaged in xenophobic posting were typically not the same individuals as those who authored misogynistic content. Given the divergent behavioral responses among users with a history of xenophobic versus misogynistic posting, one might ask whether these were groups with overlapping individuals. A simple correlation between the individual-level shares of xenophobic and misogynistic posts reveals a positive but small association of 0.041 (see Table S.14 in the Supplementary Material).

6.1.2 Do users change xenophobia for misogyny?

In this subsection, we show that one possible explanation for the observed decline in xenophobic content, alongside the stability in misogynistic content, is that users shift the group of their animosity in response to reduced anonymity. Rather than expressing less hate, users substitute one target group (e.g., immigrants) for another (e.g., women).

If some individuals use online hate to manage their frustration, they would post less of the type of hate journalists are most likely to expose. If individuals substitute xenophobic hate for misogynistic hate, it would suggest that xenophobic language is perceived as less socially accepted compared to misogynistic language.

Below, we analyze changes in the number of xenophobic and misogynistic posts at

the individual level from the pre- to the post-event period. By interacting the change in the number of xenophobic posts with the treatment variable, we capture the difference in substitution from xenophobia to misogyny between the two groups. We formally present the regression model below, in equation 3.

$$\Delta Y_i = \rho Risking\, exposure_g * \Delta X_i + \pi \Delta X_i + \sigma Risking\, exposure_g + \kappa_{ig} \tag{3}$$

ΔY_i denotes the change in the number of misogynistic posts authored by individual i, while ΔX_i represents the corresponding change in xenophobic posts. The variable *Risking exposure* is the binary indicator used above to denote membership in the treatment group. The coefficient ρ captures the relationship between changes in xenophobic and misogynistic posting behavior, specifically for users in the at-risk group. Because this analysis departs from the difference-in-differences framework, we interpret the estimated coefficients as correlations.

[Insert Table 5 here]

The correlations reported in Table 5 indicate that users in the group at risk of exposure who reduced their xenophobic posting tended to increase their misogynistic content. Specifically, the estimates suggest that, among treated users, a decrease of one xenophobic post is associated with an increase of approximately 0.35 misogynistic posts.

6.2 The effect of reduced anonymity on readable content

Turning to the unbalanced panel, we show that reduced anonymity decreases the number and the share of xenophobic posts available online for readers of the forum. The number of misogynistic posts remains stable, but the share increases.

Here, we begin by examining pre-event trends and post-event dynamic effects in standard event graphs in Figure 4.

[Insert Figure 4 here]

The figure shows coefficients from a dynamic difference-in-differences model (Angrist and Pischke 2008), estimated relative to the year before the event (Year = -1 in the figure).

First, there is no discernible pre-trend in the coefficients before the event (indicated by the vertical reference line). Second, after the event, the graph shows a significant decline in the share of published xenophobic content. For the share of misogynistic content, we observe an increase after the event, but it is not significant. These patterns support the interpretation that the observed effects stem from a decrease in anonymity, rather than from differential trends across groups.

[Insert Table 6 here]

Table 6, Panel A shows that the event significantly reduces both the number and the proportion of xenophobic entries (estimations of β from Equation (2)). Looking at misogyny in Panel B, we see that the event does not affect the number of misogynistic entries; however, we do see a significant increase in the share of misogynistic entries. This increase may be attributed to a combination of the non-significant increase in the number of misogynistic posts (Panel B) and the non-significant decrease in the total number of posts (Panel C). Overall, the results in Table 6 are broadly consistent with the behavioral patterns observed in the balanced panel analysis.[15]

To contextualize the magnitude of the observed effects, we combine three key pieces of information: (i) the average user views approximately 84 posts per day (see Section 4.2); (ii) the baseline share of xenophobic content was 29% (see Table 1); and (iii) the estimated effect of the event was a 4.1 percentage point reduction in the share of xenophobic posts (see Table 6). Taken together, these figures suggest that after the event, the average user was exposed to about 21 xenophobic posts per day, compared to 24 posts per day before the event. Applying the same approach to misogynistic content, we find that the average user encountered an estimated 21 misogynistic posts per day after the event, compared to 18 posts before the event. This indicates a rise in visible misogynistic posts despite overall reductions in posting volume.[16]

6.3 Exploring alternative explanations to the results

In this section we show that the decrease in xenophobic posts cannot be explained by i) users leaving the forum, ii) by other coinciding events, and iii) users having multiple accounts.

To assess whether users leaving the forum drives the results, we estimate the share of pre-event active users at risk of exposure who write nothing after the event, compared with users not at risk. Because all users are, by construction, active before the event, we cannot apply a difference-in-differences design to this outcome. Instead, we use the balanced panel to estimate the likelihood of exiting the platform using a cross-sectional specification that includes controls for registration date and its square (*startdate* and $startdate^2$). The analysis suggests that users at risk of exposure were 4.5 percentage points more likely to leave the forum after the event; see Table S.7 in the Supplementary Material. However, when making the users at risk of exposure more and more equal to the users not at risk by restricting the comparison to users who registered near the March 2007 cutoff, this difference in exit probability disappears (see Figure S.3d in the Supplementary Material). Our difference-in-differences estimates from the unbalanced panel are largely robust to shrinking the window around the cutoff. See Figure S.4 in the Supplementary Material. Together, these graphs suggest that the effect in the unbalanced panel is not driven by users at risk of exposure leaving the forum.

For the unbalanced panel results, we first asked whether other events during the study period may have influenced the observed changes in online hate. For instance, the results may be driven by the interval between the journalists' initial announcement and the subsequent public exposure of a few user identities. If this period drove the results, it suggests that the behavioral change stemmed not from perceived risk of exposure but from something else during that interval. To test this possibility, we re-estimated our models excluding data from the intermediate period, September 2014 to February 2015, and found that the results remained stable. Another plausible confounder is the 2014 Swedish parliamentary election, which could have affected online discourse. To rule this out, we ran an additional check by excluding the election week from our analysis. Again, the results remained qualitatively unchanged. See Table S.12 in the Supplementary Material for the corresponding estimates.

Finally, the anonymity event may have triggered a wave of new account registrations, which could have influenced our results. However, as the forum does not permit users to maintain multiple accounts, this scenario is unlikely. Indeed, we observe no spike in the number of active accounts registered after the event (see Figure S.2 in the Supplementary Material). Nevertheless, to test this more directly, in the unbalanced panel we restrict

the sample to posts authored by users who registered prior to the event (i.e., before September 2014). The results remain robust to this restriction, as expected. Additionally, for completeness, we estimate models that include controls for user registration date and its square (*startdate* and $startdate^2$), which again produce results that are substantively unchanged. Again, see Table S.9 for detailed results.

7 Discussion and conclusion

Our study demonstrates that effects of anonymity online depend on the type of prejudice expressed. We use a sudden and external reduction of anonymity on Sweden's largest online discussion forum to show that increased accountability reduced xenophobic expressions but not misogynistic expressions. These results imply that anonymity does not facilitate hate speech in a uniform way; its effects depend on the social and reputational costs of being exposed as expressing specific prejudice.

When anonymity was reduced, online xenophobia decreased but not misogyny. We also find that the decline in overall writing activity was concentrated among users who had engaged in xenophobic writing before anonymity was reduced. Moreover, our results suggest some substitution from xenophobic toward misogynistic expression among these users. Taken together, these findings indicate that the relationship between anonymity and hateful expression depends on the target group. In line with the discussions in (Álvarez-Benjumea 2025), the differentiated association with anonymity suggests that there are different anti-prejudice norms for expressing xenophobia and misogyny. Even modest behavioral changes among active contributors can substantially alter the informational environment a larger audience of readers encounters.

Our findings are consistent with the reputational theory of Ali and Bénabou (2020). If refraining from xenophobic expression provides a more informative signal of prosociality than refraining from misogynistic expression, increased accountability should thus suppress xenophobia more strongly. The substitution from xenophobic to misogynistic expression we observe also suggests that users adjusted their behavior in response to changing reputational reasons and fear of sanctions rather than changing underlying attitudes (as in the results when the U.S. government changed anti-immigration policies (Flores 2017)). If anonymity changed the users' attitudes (animosity), it would require

that they become less xenophobic and more misogynistic after the reduction in anonymity, an account we consider less plausible.

How individuals behave socially varies depending on the level of prejudice that is considered acceptable within a specific social context. An environment with fewer actual and perceived social repercussions enables increased norm-breaking through disinhibition (Suler 2004; Moore et al. 2012; Van Royen et al. 2017) and facilitates anti-social behaviors (Postmes Spears and Lea 1998). Moreover, prejudice expression depends on social context (bursztyn2020extreme; Álvarez-Benjumea 2025), peer punishment (Álvarez-Benjumea 2023; Álvarez-Benjumea 2025), and informal sanctions Álvarez-Benjumea and Winter (2018). Our results expand this literature by showing that expected punishment from the surrounding society outside the online community shapes behavior (prejudice expressions) and that responses to reduced anonymity are heterogeneous. In particular, behavioral responses depend on both the content users had produced before their risk of exposure increased and the type of prejudice being expressed. Interestingly, users responded as if the expected consequences were substantial although the actual risk of exposure was relatively low, implying that perceived reputational risk may alone alter online prejudice expressions.

The broader literature on the social regulation of prejudice or hate online has involved regulating anonymity (Siegel 2020; Froomkin 2017). Such measures include balancing participation and the expression of controversial views against the consequences of weakened accountability for harmful expression, given the influence of online hate on related hate crimes (Bursztyn Egorov and Fiorin 2020; Müller and Schwarz 2020; Müller and Schwarz 2021). One such regulation to increase accountability is using identity verification. This measure may also weaken the protection of vulnerable writers in both democratic and authoritarian countries (Froomkin 2017). Decreased anonymity could lead to less hateful language and limit discussions of controversial topics (Zhang and Kizilcec 2014). For example, restrictions introduced in China in 2012 requiring greater identification on social media were followed by increased self-censorship in political discussions (Fu Chan and Chau 2013). Given our results, understanding how individuals expressing different forms of prejudice respond to changes in anonymity is therefore important for evaluating measures taken to improve online discussion environments.

For all studies based on a single natural experiment, questions of external validity

remain. Although our empirical setting is Sweden, the mechanisms are likely relevant to other similar countries. The central factor is the variation in social cost attached to expressing different forms of prejudice. Such costs will differ in magnitude across countries, online platforms, and historical periods, but the underlying mechanism is likely similar.

Future studies should explore the normative environments of xenophobic and misogynistic expressions and possibilities of expressing controversial opinions across countries and historical time periods. In particular, such studies could distinguish more directly between changes in underlying attitudes and changes in the willingness to publicly express those attitudes. Moreover, we need more knowledge about when policies intended to increase accountability reduce harmful expression without producing broader chilling effects on legitimate public discussion. Understanding these differences is essential for explaining how anonymity shapes online behavior and for evaluating attempts to balance open expression with accountability in digital public spheres.

The central message from our study is that anonymity should not be treated as a uniform driver of online hate. In our setting, reduced anonymity suppresses xenophobic writing in our setting but does not have the same effect on misogyny, and some users appear to substitute one form of hateful expression for another (eklund2020beyond). We show that drivers of the results are the meaning attached to the social norms and reputational costs associated with various types of prejudice. Understanding those differences is essential for explaining online behavior and for designing interventions that seek to reduce harmful expression while preserving open public discussion.

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Notes

1. Although there is an increasing body of literature on using text as data in political science, sociology, and economics (grimmer2013text; evans2016machine; gentzkow2019text), there remains no consensus on methods or models for automatically detecting xenophobia and misogyny (davidson2017automated; badjatiya2017deep). Also, our setup is thus a typical example in which the canonical difference-in-differences method applies, with one group receiving treatment once and going forward and a control group that is never exposed. A relatively new literature expands the methodology for difference-in-differences approaches to settings where, for example, the treatment timing varies, so-called staggered models. These methods do not apply in our case, since we only have two groups without staggered treatment introduction.

2. Discussions on Flashback are available at https://www.flashback.org/, and on Reddit at https://www.reddit.com/.

3. In 2002, a Swedish court banned the founder of Flashback from running the business, leading the founder to run the business abroad.

4. The thread is called "Researchgruppen påstår att de har Flashbacks databas (del 2)" and can be found at `https://www.flashback.org/t2526334p1`.

5. Authors' translation of post no. 30 written on 2015-02-16, 18:45 within the thread. Access `https://www.flashback.org/t2526334p1`.

6. Randomization was implemented at the thread level to assess whether the initial content influenced subsequent posts and to determine whether debates emerged that critiqued previous contributions. Analyzing posts before and after the event ensured a consistent linguistic framework across the entire sample period. Training the model exclusively on pre-event data, for instance, could have led to an overestimation of the decline in xenophobic or misogynistic language if linguistic patterns shifted as a consequence of the event.

7. The Supplementary Material provides additional details on the machine learning models and addresses potential measurement errors associated with their application. See sections S.2, S.2.1 and S.3.

8. Alexa. 2017. "Internet Rankings," Accessed January 1, 2017. `https://www.alexa.com/siteinfo/flashback.org`

9. One potential concern relates to the estimation of standard errors in a difference-in-differences setting with only one treatment group and one control group (bertrand2004much; donald2007inference; conley2011inference; Pettersson-Lidbom and Skogman Thoursie 2013). We estimate Equation (2) using the Newey–West estimator with four lags to address this issue. This approach adjusts for both autocorrelation within groups and serial correlation over time, yielding standard errors that are more robust to the known limitations of inference in aggregated DiD settings.

10. For completeness, Table S.8 in the Supplementary Material provides the corresponding estimates with and without these control variables.

11. The main drawback of the balanced panel is that we can't test the parallel trend assumption using a standard event graph due to the two-time-period construction of the data. As an additional robustness check, we instead exploited variation in user registration dates to examine whether our results hold in increasingly comparable samples. Specifically, we compared the central estimate with estimates from models restricted to users who registered within progressively narrower time windows, from five years to one year, around the March 2007 cutoff. As the window narrows, the at-risk and control groups become more similar in observed and unobserved characteristics, including pre-existing trends. Under the parallel trends assumption, we expect coefficient estimates to remain stable while confidence intervals widen due to the reduced sample size. Unlike models that rely on functional form assumptions (e.g., linear or quadratic controls for registration date), this approach avoids parametric assumptions by design. Figure S.3 in the Supplementary Material displays the resulting coefficients and 95% confidence intervals for four outcomes: the number of xenophobic posts, the number of misogynistic posts, the total number of posts, and the probability of exiting the forum. Our estimated effects remain remarkably stable across the first three panels.

12. These estimates are robust to model specification: the results remain unchanged when the registration date control functions are excluded (see Table S.9 in the Supplementary Material).

13. The individuals in the balanced panel may constitute a selected subset of the broader user population represented in the unbalanced panel. To assess potential selection bias, we compared users in the balanced panel to those who ceased posting after the event. The comparison reveals striking similarities between the two groups. Specifically, users in both samples exhibited comparable pre-event levels of xenophobia and misogyny, as well as similar average registration dates (see Table S.3 in the Supplemental Material).

14. The results presented in Table 4 are robust to both including and excluding controls for user start date; see Table S.11 in the Supplementary Material.

15. Estimates from a difference-in-differences model applied to the manually coded random sample by the research assistant yield qualitatively similar results (see Table S.6 in the Supplementary Material).

16. Notably, in this back-of-the-envelope exercise, we assume that all users are affected similarly by the event.

Table 1: Summary statistics

	Mean	SD
Panel A: Balanced panel of individuals		
No. entries	58.456	236.904
No. xenophobic	18.728	79.636
No. misogynistic	14.686	95.996
Share xenophobic	0.294	0.289
Share misogynistic	0.213	0.262
N	22318	
Users	11159	
Panel B: Unbalanced panel of posts		
Share xenophobic	0.329	0.470
Share misygynistic	0.258	0.438
N	1984224	
Users	48672	

The table presents the summary statistics
for the balanced (Panel A) and unbalanced (Panel B) panels.
The variables are based on predictions from fine tuned BERT models.

Table 2: Pre-event differences between those at risk and not at risk of exposure

Panel A: Xenphobia		
	No. xenophobic	Share xenophobic
Risk exposure	3.260	0.023
	(6.641)	(0.014)
Startdate linear control	Yes	Yes
Startdate quadratic control	Yes	Yes
Observations	11159	11159
Panel B: Misogyny		
	No. misogynistic	Share misogynistic
Risk exposure	−2.645	−0.011
	(3.673)	(0.012)
Startdate linear control	Yes	Yes
Startdate quadratic control	Yes	Yes
Observations	11159	11159
Panel C: No. entries		
	No. entries	
Risk exposure	−6.813	
	(15.147)	
Startdate linear control	Yes	
Startdate quadratic control	Yes	
Observations	11159	

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until September 10th 2014. It thus covers the the period in our data before the event. Standard errors clustered at the user level. The dependent variable is the number of individual xenophobic entries and the share of xenophobic entries prior to the event in Panel A, the number of individual misogynistic entries and the share of misogynistic entries in Panel B and the number of individual entries in panel C. All regressions include controls for user start, date, user start date squared and an intercept.

Table 3: Impact of a decrease in anonymity on xenophobic and misogynistic writings

Panel A: Xenophobia		
	No. xenophobic	Share xenophobic
Post*Risk exposure	-7.529^{**}	0.012
	(2.673)	(0.009)
Startdate linear control	Yes	Yes
Startdate quadratic control	Yes	Yes
Observations	22318	22318
Panel B: Misogyny		
	No. misogynistic	Share misogynistic
Post*Risk exposure	-1.082	-0.003
	(1.575)	(0.008)
Startdate linear control	Yes	Yes
Startdate quadratic control	Yes	Yes
Observations	22318	22318
Panel C: No. entries		
	No. entries	
Post*Risk exposure	-14.711^{**}	
	(5.603)	
Startdate linear control	Yes	
Startdate quadratic control	Yes	
Observations	22318	

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until September 10th 2014. It thus covers the the period in our data before the event. Standard errors clustered at the user level. The dependent variable is the number of individual xenophobic entries and the share of xenophobic entries prior to the event in Panel A, the number of individual misogynistic entries and the share of misogynistic entries in Panel B and the number of individual entries in panel C. All regressions include controls for user start date, user start date squared, a dummy for if the user risked exposure, a dummy for if the time period is post the event and an intercept.

Table 4: Impact of a decrease in anonymity on number of written entries, by level of hate prior to event

	Dependent variable:			
	No. entries			
	(1)	(2)	(3)	(4)
Post*Risked exposure	−1.346	−28.662**	−13.133*	−17.714
	(5.658)	(9.901)	(6.330)	(9.594)
Sample	Xenophobia below median	Xenophobia above median	Misogyny below median	Misogyny above median
Start date linear control	Yes	Yes	Yes	Yes
Start date quadratic control	Yes	Yes	Yes	Yes
Observations	10900	11418	11158	11160

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until December 31st 2016. Standard errors clustered at the user level. Each column uses the data in either the bottom 2/3 or top 1/3 of the distribution in individual share xenophobia and share misogyny in the pre-treatment period, respectively. Outcome variable is individual number of entries in the pre- and post-period. All regressions include controls for user start date, user start date squared, a dummy for if the user risked exposure, a dummy for if the time period is post the event and an intercept.

Table 5: Estimation of the substitution between xenophobia and misogynism

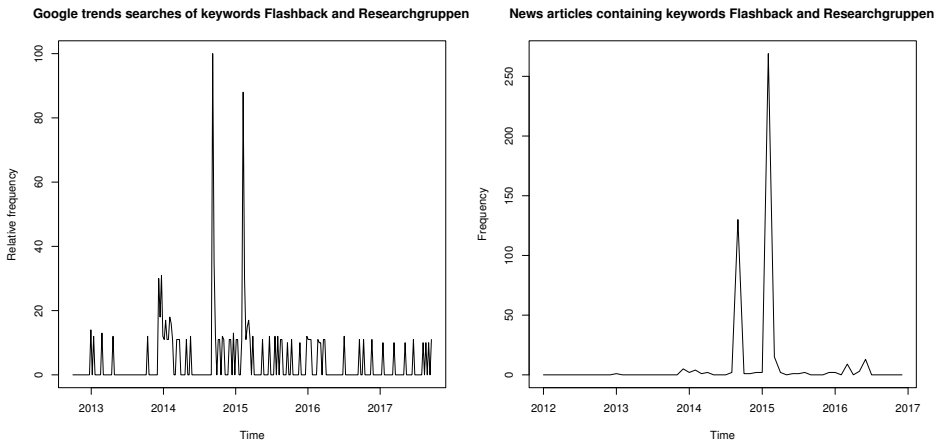
	Dependent variable:		
	ΔNo. misogyny	ΔNo. misogyny	ΔNo. misogyny
	(1)	(2)	(3)
Risk exposure*ΔNo. xenophobic	-0.351^{**}	-0.350^{**}	-0.350^{**}
	(0.129)	(0.128)	(0.128)
ΔNo. xenophobic	0.687^{***}	0.686^{***}	0.686^{***}
	(0.126)	(0.126)	(0.126)
Risk exposure	1.479	3.459	2.261
	(1.190)	(2.140)	(2.446)
Start date linear control	No	Yes	Yes
Start date quadratic control	No	No	Yes
Observations	11159	11159	11159

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until December 31st 2016. Standard errors clustered at the user level. Outcome variable is number of misogynistic entries at the individual level.

Table 6: Impact of a decrease in anonymity on xenophobic and misogynistic writings, unbalanced panel

Panel A: Xenophobia		
	No. xenophobic	Share xenophobic
Post	-456.679^{*}	-0.046^{***}
	(200.007)	(0.007)
Observations	261	261
Panel B: Misogyny		
	No. misogynistic	Share misogynistic
Post	49.102	0.035^{***}
	(110.265)	(0.007)
Observations	261	261
Panel C: No. entries		
	No. entries	
Post	-763.480	
	(420.342)	
Observations	261	

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until December 31st 2016. The data is collapsed at the weekly level per treatment/control group, between which the difference is computed, giving us a single time series. To this, Newey-West standard errors are computed using 4 lags. Thus, Post represent the difference in difference estimates. The outcomes are the difference in the number and share of xenophobic and misogynistic entries and the total number of entries per week between the group at risk of exposure and those not at risk.



(a) (b)

Figure 1: Google searches and news articles relating to the event

The left graph displays data from Google Trends for the weekly relative search frequencies of the terms Flashback and Researchgruppen between 2013 and 2017. The relative search frequency shows a spike if there are at least 50 searches, and all of the spikes are relative to the largest spike, which reaches 100 in the figure. Thus, a spike reaching 80 implies that the search terms reach 80% of the highest search frequency. The right graph displays a graph similar to the monthly number of news articles in Sweden that contain the words "Flashback" and "Researchgruppen". The data on the number of news articles were obtained through searches in the Mediearkivet database (https://www.retriever.se/, accessed using a login from Stockholm University on July 4, 2018).

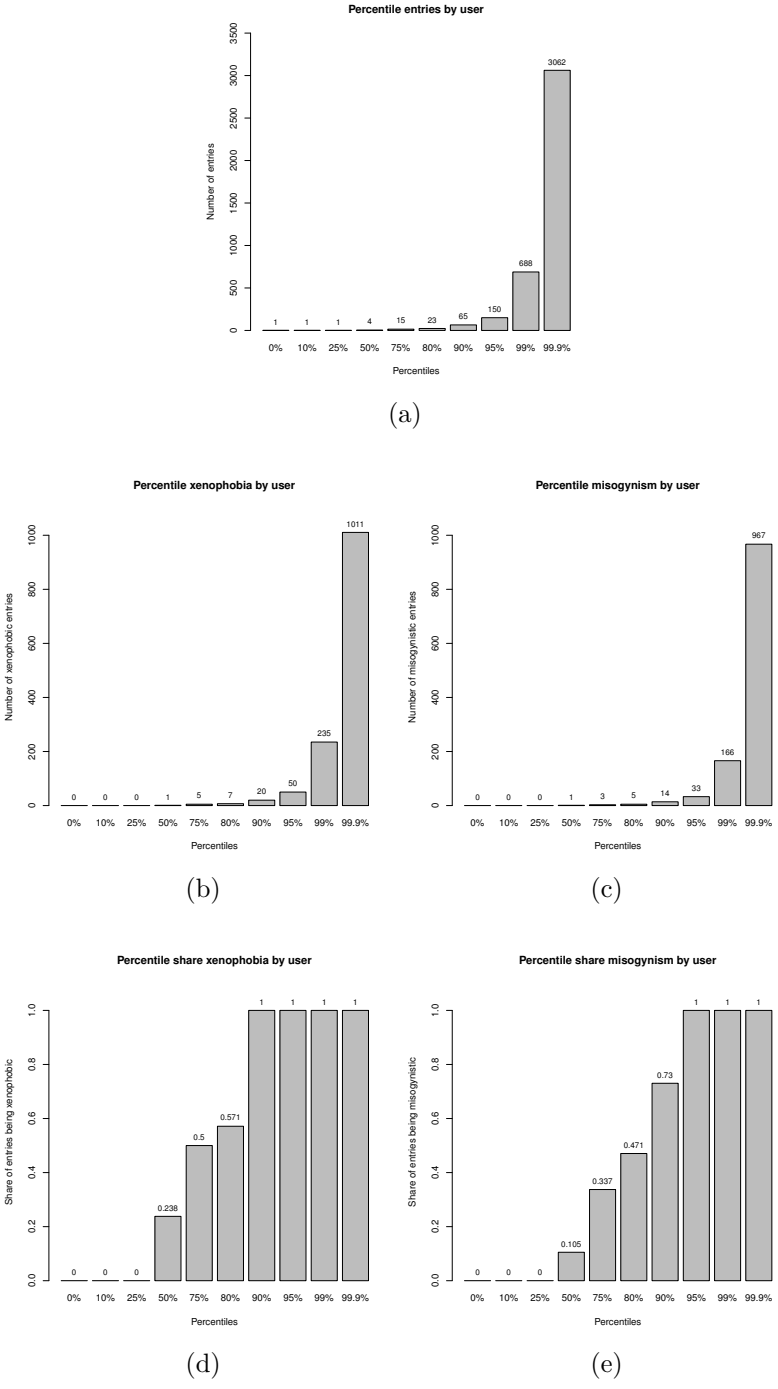
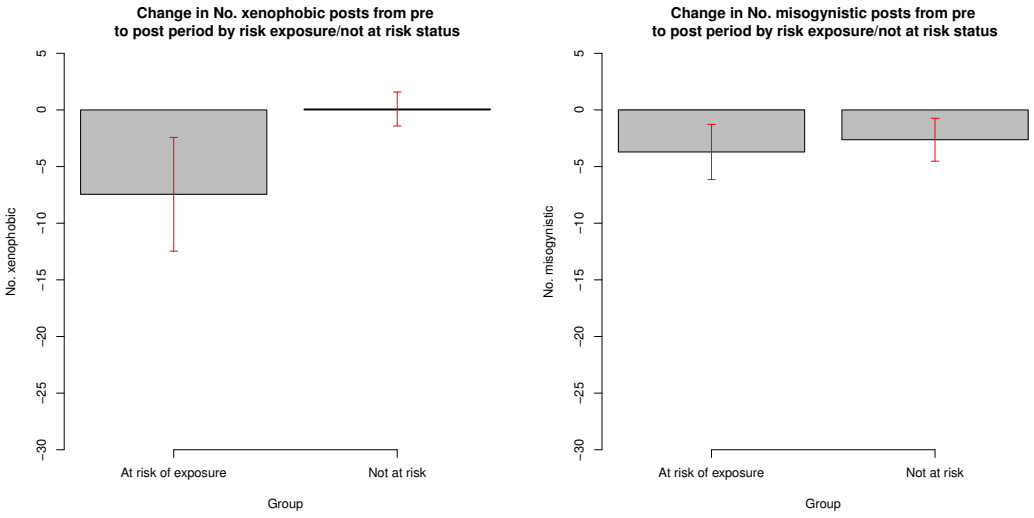


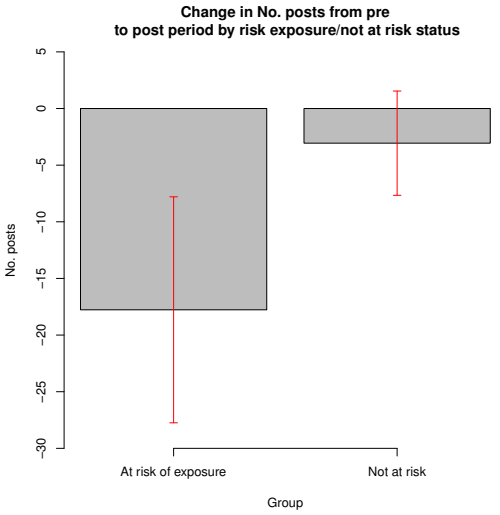
Figure 2: Distribution of number of posts, and number and share of xenophobia and misogyny

The top part presents the distribution of different percentiles of the total number of entries users wrote in our data. The second part from the top presents the distribution of different percentiles of the total number of xenophobic entries users wrote in our data. The middle part presents the distribution of different percentiles of users' share of written xenophobic entries in our data. The second-to-last part presents the distribution of different percentiles of the total number of misogynistic entries users wrote in our data. The last part presents the distribution of different percentiles of users' share of written misogynistic entries in our data.



(a)

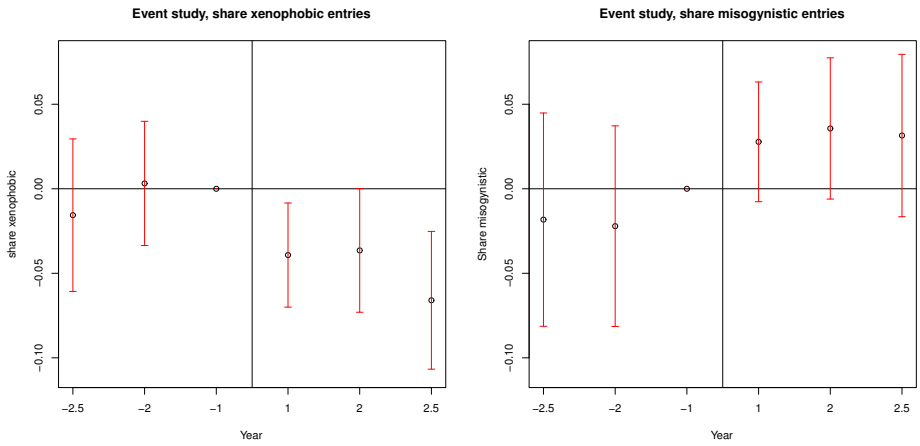
(b)



(c)

Figure 3: Change in content from pre to post period by exposed/not

The bars demonstrates coefficientsizes from separate regressions per at risk and not at risk of exposure on the post dummy variable along with the 95% confidence intervals. The period used in the regressions is from January 1, 2012, to December 31, 2016. Standard errors are clustered at the user level.



(a) (b)

Figure 4: Event study: Difference-in-difference before and after the event, unbalanced panel

The coefficients display the interaction terms between the risk exposure dummy and time dummies along with the 95% confidence intervals from a dynamic difference-in-difference model in the unbalanced panel. The time dummies are defined based on the time from the initial revelation on September 10, 2014, and thus do not correspond to calendar years. The period used in the regressions is from January 1, 2012, to December 31, 2016. Standard errors are clustered at the user level.

Supplementary Material: Misogyny and Xenophobia Online: A Matter of Anonymity

September 17, 2026

Abstract

In this supplementary material, we demonstrate how machine learning can be used to obtain measures of xenophobia and misogynism for all of the available data from an online forum. We also show how well the models predict the different outcomes, if measurement error in the predicted measures from the machine learning models cause bias in the difference-in-difference estimates and how to obtain unbiased estimates. Finally, we provide additional tables and robustness analysis.

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S.1 The data from the research assistant

The data used in this study consist of text-based messages (referred to as posts or entries) collected from the Flashback discussion forum. Using a custom-built Python script, we scraped all posts from three subforums—feminism, domestic politics, and immigration—from the inception of each forum until January 2017. To analyze these data, we required a method for identifying xenophobic and misogynistic content in written text. Defining such content is inherently challenging, as expressions of xenophobia and misogynism are context-dependent and often subtle. To address this, we employed a research assistant (henceforth RA) to manually classify posts based on a detailed set of instructions. These instructions, provided in Section S.7 of this Appendix, offered clear guidelines, but the ultimate judgment in classification was left to the RA's discretion. Importantly, the RA was explicitly instructed to be conservative in her classifications. This means that, if anything, our coding likely underestimates the prevalence of xenophobia and misogyny in the dataset. Furthermore, the RA was unaware of our research design and our specific research question.

We began by randomly selecting 100 threads from each of the three subforums. Within each thread, the research assistant (RA) classified the first twelve and the last five posts. Randomization was conducted at the thread level to allow us to examine whether an initial instance of xenophobic or misogynistic language influenced the tone of subsequent posts—that is, whether early expressions of hate generated further hateful content. We were also interested in identifying whether a discursive counter-response occurred, whereby earlier posts were contested or criticized by other users. Using our developed instructions, the RA classified each post according to whether it contained hateful content, threatening language, or aggressive language, and identified the target group—women and feminists, foreign residents, or others. In addition, the RA classified whether the post affirmed or challenged the claims made in preceding posts, whether it expressed support for or opposition to a specific political party, and whether it employed "us versus them" rhetoric. The classification definitions were informed by previous research on linguistic indicators of online hate and polarization (Cohen et al. 2014). The final dataset consisted of 4,040 manually classified posts, approximately evenly distributed across the three subforums.

S.2 Prediction of racism and sexism

To create measures of xenophobia and misogynism, we needed some form of baseline input/numeration for parts of the data that we could use to train and evaluate a prediction model. As described in Section S.1, we choose to employ a research assistant (RA) to manually classify posts from a randomly selected subset of threads. We used this manually coded data to train two machine learning models designed to predict the presence of xenophobic and misogynistic content. Following established methodological practice (James et al. 2013), we randomly divided the data into a training set (70 percent of the sample) and a test set (the remaining 30 percent).

We used the training data to fine-tune a Bidirectional Encoder Representations from Transformers (BERT) model to predict xenophobic and misogynistic content. This approach enabled us to leverage BERT's contextual understanding of word relationships and sentence structures, while adapting the model to the specific classification tasks in our study. In practical terms, fine-tuning involves reweighting the parameters of the pre-trained BERT model to optimize its performance for our prediction objectives. Previous research has shown that BERT models fine-tuned for domain-specific classification tasks can match or even exceed human annotators in predictive accuracy (Do and Ollion and Shen 2024). The primary advantage of using a fine-tuned BERT model over simpler machine learning algorithms thus lies in its superior accuracy, as demonstrated in Section S.2.1. However, a key limitation of this approach is interpretability. The BERT model functions as a black box, making it difficult to identify the precise features or linguistic cues that led to a particular post being classified as xenophobic or misogynistic.

Earlier, we employed the more simple logistic lasso algorithm to predict xenophobic and misogynistic content. The key advantage of the logistic lasso—unlike the BERT model—is its high interpretability. However, this comes at the cost of lower predictive accuracy. Table S.10 presents the main results from the balanced panel using predictions from the logistic lasso model. These results can be directly compared to those based on the BERT model predictions reported in Table S.9. As outlined in our theoretical discussion in Section S.3, the logistic lasso model introduces greater measurement error, which attenuates the estimates. Consistent with this theoretical prediction, the results based on logistic lasso predictions are directionally similar but generally weaker than those obtained using the BERT model, though still statistically significant. Although we do not present the full set of results using logistic lasso predictions in this version of the paper, we note that the overall conclusions remain unchanged. In general, we

observe stronger effects with larger coefficients in absolute values when using BERT-based predictions, relative to those derived from the logistic lasso. Previous drafts of the paper are available upon request to to demonstrate this claim.

S.2.1 Evaluation of the predictions from the BERT model

In order to understand if our prediction models accurately predicted xenophobia and misogynism, we needed to evaluate their performance. Thus, after fine-tuning the two models to predict xenophobic and misogynistic content, we evaluated their predictive performance using the designated test set. Specifically, we applied the models to the observations in the test data and compared the resulting predictions to the original classifications provided by the RA. This allowed us to compute the false negative and false positive rates across different probability cut-off values.

We first assessed model performance using the standard Receiver Operating Characteristic (ROC) curve, presented in Figure S.1, and computed the Area Under the Curve (AUC) as a general metric of classification accuracy. The ROC curve illustrates the trade-off between the false positive rate and the true positive rate (i.e., 1− the false negative rate), while the AUC represents the proportion of the area under the ROC curve relative to the total possible area. AUC values range from 0 to 1, with values below 0.5 indicating worse-than-random prediction and values closer to 1 reflecting stronger model performance. The main advantage of the ROC curve and AUC is that they both provide interpretable, threshold-independent measures of model quality. Figure S.1a displays the ROC curve for misogyny predictions, while Figure S.1b presents the corresponding curve for xenophobia. Both models perform well, with an AUC of 0.91 for misogyny and 0.89 for xenophobia.

[Figure S.1 here]

To classify posts as xenophobic or misogynistic in the full dataset, we needed to set a threshold probability above which a post would be labeled as containing such content. As we demonstrate in Section S.3, the attenuation bias in difference-in-differences estimates with a binary outcome variables is minimized when the sum of the false positive rate and the false negative rate is minimized. Accordingly, we selected the cut-off value that minimizes this sum, thereby reducing the influence of measurement error on our

coefficient estimates.

Once the classification threshold had been selected, we applied it to the model predictions and evaluated performance by comparing them to the RA's manual classifications in the test set. This comparison is summarized using confusion matrices—2×2 tables with the RA's actual classifications on the y-axis and the model's predicted classifications on the x-axis—for each outcome variable. Table S.1 presents the confusion matrix for xenophobia, while Table S.2 provides the corresponding results for misogyny.

Starting with xenophobia, Table S.1 shows that, out of 1,212 observations in the test set, 91 were correctly classified as xenophobic and 888 were correctly classified as non-xenophobic. In contrast, 213 posts were classified as xenophobic although they were not (false positives), and 20 posts were classified as non-xenophobic despite being xenophobic (false negatives). These figures yield a false negative rate of $\frac{20}{20+91} \approx 0.18$ and a false positive rate of $\frac{213}{213+888} \approx 0.19$. The overall accuracy of the model in predicting xenophobic content is thus $\frac{91+888}{1212} \approx 0.81$.

Turning to misogyny, Table S.2 shows a similar pattern. In this case, 52 observations were correctly classified as misogynistic and 838 as non-misogynistic. 320 were in-turn false positives—classified as misogynistic though they were not—and 2 were false negatives—classified as non-misogynistic despite being misogynistic. From these values, we compute a false negative rate of $\frac{2}{2+52} \approx 0.037$ and a false positive rate of $\frac{320}{320+838} \approx 0.28$. The overall accuracy for the misogyny model is therefore $\frac{52+838}{1212} \approx 0.73$.

In sum, both models demonstrate strong predictive performance. The xenophobia model achieves slightly higher precision, while the misogyny model exhibits a lower false negative rate and a slightly higher AUC-score. Thus we conclude that our automated classification procedure based on the RA-coded training data is highly reliable.

[Table S.1 and S.2 here]

S.3 Measurement Error in the dependent variables

Even though our prediction models according to our evaluations work well and have high precision, the fact that we are using algorithms to predict xenophobia and misogynism necessarily means that we will have some measurement error. In this section, we outline a simple framework for understanding the role of measurement error in our empirical setting. In the main analysis, we use two types of outcome variables: (1) binary indicators of xenophobic and misogynistic content based on predictions from our BERT models, and (2) the (standardized) counts of xenophobic, misogynistic, and total posts. The total number of entries is measured directly and is therefore unaffected by measurement error. In contrast, the variables capturing xenophobic and misogynistic content rely on predictions and are thus subject to some degree of measurement error.

Our difference-in-differences (DiD) design mitigates certain types of measurement error. Specifically, any error that is constant over time but varies between the treatment and control groups will be absorbed by the group fixed effects. Similarly, any temporal trend in measurement error that affects both groups equally will be differenced out through the time fixed effects. Of greater concern are forms of measurement error that vary both across groups and over time—such as systematic shifts in language use (e.g., increased reliance on euphemisms) following the exposure event. However, our classification models were trained on manually coded data produced by a research assistant (RA) who was blind to the research design. The RA coded posts from both exposed and non-exposed users and from both the pre- and post-event periods. As a result, any systematic bias in classification should be unrelated to treatment status or time. Formally, this implies a parallel trends assumption in measurement error across groups. As expected, we observe no evidence of differential trends in our event study figures (see the main text), suggesting that measurement error trends are likely stable across treatment and control groups.

Despite these safeguards, some caveats remain. For continuous outcome variables—such as the (standardized) number of xenophobic or misogynistic posts—classical measurement error will inflate the standard errors but leave the coefficient estimates unbiased. However, for binary outcome variables, measurement error introduces attenuation bias. That is, false positives (non-xenophobic or non-misogynistic posts misclassified as such) and false negatives (xenophobic or misogynistic posts misclassified as neutral) will bias the estimated treatment effects toward zero. We show this formally using a simple framework

below, using xenophobia as our example. Note that the analysis would be the same for misogynism, however.

Suppose that there is some unobserved true amount of xenophobia in a post, denoted by $Y^\star$. Furthermore, a post is deemed xenophobic ($Y = 1$) by a representative individual if $Y^\star$ exceeds some unobserved threshold c;

$$Y = \begin{cases} 1, & \text{if } Y^\star \geq c \\ 0, & \text{if } Y^\star < c \end{cases} \tag{1}$$

In other words, a standard binary outcome model. In addition, we are interested in the effect of a binary treatment indicator x, taking the value of 1 in case of treatment (risking exposure) and zero otherwise (not at risk). Thus, we have a standard binary variable as an outcome in this setting, and we are interested in the relationship

$$Y = \alpha + \beta x + \varepsilon. \tag{2}$$

Now, let us introduce measurement error. In the paper, we define xenophobia as expressions covering prejudices involving ethnicity and/or race. It is however not feasible for us to obtain this measure; it is unknown to us by nature. Thus, we do not observe Y either, due to what we can call the feasibility constraint. Rather, we observe the coding of our RA, $\tilde{Y}$, given by:

$$\tilde{Y} = \begin{cases} Y, & \text{with probability } 1 - q_{01} - q_{10} \\ 1, & \text{if } Y = 0 \text{ with probability } q_{01} \\ 0, & \text{if } Y = 1 \text{ with probability } q_{10} \end{cases} \tag{3}$$

In other words, with probability q_{10}, the RA classify a xenophobic entry as non-xenophobic, and with probability q_{01}, the RA classify a non-xenophobic entry as xenophobic. By construction, the relationship between Y and $\tilde{Y}$ is less than one since either $\tilde{Y} = Y$, i.e., a one-to-one relationship, or $\tilde{Y} = 1 - Y$, i.e., a negative measurement error. A regression of $\tilde{Y}$ on Y of the form:

$$\tilde{Y} = \theta_0 + \theta_1 Y + \varphi \tag{4}$$

will thus give us $\theta_0 > 0$ and $\theta_1 < 1$. We hence have a mean reversion measurement error on the left-

hand side in our regressions, which will attenuate our estimate toward zero. This error hinges on the assumption that $\theta_1 > 0$, which means that there is a positive, although imperfect, relationship between $\tilde{Y}$, the RAs measure, and Y, the unknown "true" measure of xenophobia. In other words, our RA must outperform random chance. If, however, the RA does not outperform random chance so that $\theta_1 < 0$, our estimates will have reversed signs, though we find this extremely unlikely.

The feasible population regression function in our setting is thus

$$\tilde{Y} = \tilde{\alpha} + \tilde{\beta}x + \tilde{\varepsilon}. \tag{5}$$

Furthermore, in the main paper, we also use the predicted values from the BERT model as binary outcome variables in difference-in-difference specifications. We consider this use an additional misclassification problem stemming from the machine learning model. Thus, we have:

$$\bar{Y} = \begin{cases} \tilde{Y}, & \text{with probability } 1 - p_{01} - p_{10} \\ 1, & \text{if } \tilde{Y} = 0 \text{ with probability } p_{01} \\ 0, & \text{if } \tilde{Y} = 1 \text{ with probability } p_{10} \end{cases} \tag{6}$$

further attenuating our estimate toward zero. When we use predicted values as outcomes, the actual regressions that we run are thus of the form

$$\bar{Y} = \bar{\alpha} + \bar{\beta}x + \bar{\varepsilon}. \tag{7}$$

We can then, however, note that

$$\bar{\beta} = P[\bar{Y} = 1|x = 1] - P[\bar{Y} = 1|x = 0] \tag{8}$$

$$= (1 - p_{01} - p_{10})(P[\tilde{Y} = 1|x = 1] - P[\tilde{Y} = 1|x = 0])$$

$$= (1 - p_{01} - p_{10})\tilde{\beta} \Rightarrow$$

$$\tilde{\beta} = \frac{\bar{\beta}}{(1 - p_{01} - p_{10})} \tag{9}$$

Now, since we evaluate the performance of our prediction model with the test set in section S.2.1, we actually obtain estimates of both p_{01} and p_{10}, which we can use to recover $\tilde{\beta}$ from our obtained $\bar{\beta}$. In other words, we can take the measurement error from the prediction part into account when we compute our estimates, but we cannot account for any potential faulty classification by our RA. Thus, we cannot obtain β, only $\tilde{\beta}$, which we compute in section S.5. Finally, it is worth noting that Equation 9 implies that, since we minimize the sum of the false positive rate and false negative rate in section S.2, we also minimize the attenuation bias in our difference-in-difference estimates for the binary outcome variables. Finally, it is worth pointing out again that this only applies to the binary outcome variables, i.e. the share of xenophobic/misogynistic posts.

S.4 Simple differences-in-differences using the RA data

As noted in the measurement error section above, the data from the RA, although comprising a small sample, does not contain any measurement error stemming from the prediction algorithms. Thus, we also estimate simple difference-in-difference specifications using the raw annotated data from the RA.

Table S.5 presents the summary statistics for the main variables from the RA data, but breaks down the statistics by exposed/non-exposed and also before and after the revelation that journalists had access to their identities.[1] Since we do not have any prediction problems for this data, we can also look at the more hard to predict outcomes; hate directed at anyone, threats and aggressive writing. From this table, we can compute the simple difference-in-difference (DD) estimate for hate as $(0.14-0.21)-(0.28-0.23) = -0.12$, while the same estimate for xenophobia is $(0.03 - 0.08) - (0.11 - 0.10) = -0.06$, and for misogynism it is $(0.04 - 0.03) - (0.09 - 0.06) = -0.02$. Furthermore, for threats, the estimate is $(0.00 - 0.01) - (0.01 - 0.01) = -0.01$, and for aggression, it is $(0.41 - 0.56) - (0.67 - 0.63) = -0.19$.

[Table S.5 here]

Table S.6 provides the coefficients and standard errors for the corresponding regressions. Unsurprisingly, the estimated effects are all of the same magnitudes as the computations that we just performed, although only the estimates for hate, xenophobia and aggression are significant at the 5 percent level. It

[1] See section in the main paper for more details.

is worth noting that the main coefficient for threats come remarkably close to being significant at the 5 percent level (p-value of 0.066), although a single threatening post in the group risking exposure in the post period would be sufficient to offset the negative coefficient completely. We are thus under-powered to say anything with statistical precision about threats, but the coefficient points in the same direction as for the other outcomes. The interpretation of the effect for xenophobia is that exposed individuals are 6.4 percentage points less likely to write a xenophobic post after the event on September 10 2014. Correspondingly, exposed individuals are 12.8 percentage points less likely to write a hateful comment directed at anyone, and 18.7 percentage points less likely to write something containing aggression.

[Table S.6 here]

S.5 Example of adjusting coefficients using the false positive and false negative rates

In section S.3 we demonstrated how the measurement error affected the bias in our main estimates for the binary outcome variables, and argued that we could obtain the unbiased coefficient using equation 9 and the estimates from section S.2.1. Let us now provide some simple examples of how we can use these theoretical results with our empirical estimates from our prediction evaluations to correct our coefficients and in essence remove the measurement error.

In section S.2.1, for xenophobia we find a false positive rate of 0.19 and a false negative rate of 0.18. Adjusting our estimate for xenophobia from Table 8 in the main paper, we obtain a corrected estimate of $-0.041/(1-0.19-0.18) = -0.065$. Similarly, for misogyny, we have a false positive rate of 0.28, false negative rate of 0.037. Thus, adjusting our estimate for sexism from Table 8 gives us $0.035/(1-0.28-0.037) = 0.051$.

We can now compare these estimates with those from the RA-sample in table S.6, which are not subject to the attenuation bias from the prediction models. We find an adjusted estimate from the full sample for racism of -0.065, which is very close to the RA-sample estimate of -0.064. However, when

we look at the estimates for misogyny, we see some discrepancies, as the adjusted full sample estimate is 0.051, while the RA-sample based estimate is an insignificant −0.024.

S.6 Behind the event: How could the journalists access the identity of the users and how did the owner verify those at risk?

A common question that arises is how the journalists could obtain the identities for the early registered users when Flashback was supposed to be securely anonymous. In this section we thus outline the process that the journalists used to obtain the identities behind the early registered accounts at Flashback.

The group of journalists originally had obtained a list, a so called dump, of e-mail addresses that were connected to usernames from Flashback. The list was however from a few years back, which is why only users registered before March 2007 were in it. To this list, the users' identities were added by matching the e-mail addresses from Flashback to e-mail addresses from other leaked Internet sites, such as CDON.com, from which other data, such as Swedish social security numbers and names, could be collected. Thus, the journalists could link usernames to identities of individuals with accounts registered at Flashback before March 2007.

The journalists had a history of publicly exposing individuals writing anonymous xenophobic comments online, making the threat of future exposures credible. A spreadsheet-file containing the email addresses and identities that the journalists had access to was subsequently posted publicly online, but without the associated usernames from Flashback. The owner of Flashback could however see that the rows of the file were sorted alphabetically based on the usernames, implying that the journalists had access to the usernames as well. Furthermore, the owner could see that the accounts that were in the file were registered before March 2007. The owner of Flashback therefore declared in a thread[2] that those who had registered before March 2007 were at risk of being exposed. Said thread have to this day been read more than 300,000 times, thus making it clear that many users wanted to know if they were at risk. Several users expressed their disappointment and outrage that this had happened in the thread, and declared that they may leave Flashback as a consequence.

[2]See "Researchgruppen påstår att de har Flashbacks databas (del 2)", can be found at `https://www.flashback.org/t2526334p1`.

S.7 Instructions to the RA

Instructions to Research Assistant Spring 2017.

The population we investigate is from the Internet forum Flashback. We have scraped text from the following three sub-forums; immigration, feminism and domestic policy. We have drawn a random sample of 100 threads from each forum. Each thread and each post has an id-number. We want you to code 12 posts starting from the beginning of the thread and 5 posts starting from the end of the thread.

The unit of coding is the post. Please read the full post. You will receive the threads and posts in an Excel sheet, where we want you to insert your classifications. Below you can find descriptions of how we want you to classify the posts.

Start with 2 threads and after this we can meet to discuss the progress before you proceed.

	Responds to Svarar på	
	0 = Doesn't seem to respond to any particular post 999 = Response to several posts from several authors [tomt] = Responds to a post that's not in the sample	*The value noted here is the id-number of the post to which the writer seems to respond.*
1	Questioning Ifrågasättande	
	0 = Neither nor 1 = Affirmative 2 = Nuancing 3 = Questioning	*- If the post **quotes another post** the coding relates to the quoted post.* *- If the post **doesn't contain a quote** the coding relates to the first post.* *- The **first post is always coded** as "neither-nor"*
2	Understanding Förståelse	
	0 = No 1 = Yes	If the writer shows understanding of the thoughts and intentions expressed in an earlier post, the coding should be "yes". **Regardless of whether the writer agrees or not.**
3	Party politics positive. *Does the post express a opinion in favor of any party or coalition of parties or Feminist Initiative?*	
	0 = No, not positive to any coalition of parties 1 = Yes, the red-green coalition 2 = Yes, the liberal-conservative coalition 3 = Yes, the Sweden Democrats 4 = Feminist Initiative 5 = Feminist initiative and the Left Party 6 = The seven traditional parties 7 = Sweden Democrats and the Right.	***Only to be coded "yes" if it is obvious,** e.g. when the parties or their representatives are mentioned, either explicitly or through paraphrases*
4	Party politics negative. *Does the post express a **negative** opinion of any party or coalition of parties or Feminist Initiative?*	
	0 = No, not positive to any coalition of parties 1 = Yes, the red-green coalition 2 = Yes, the liberal-conservative coalition 3 = Yes, the Sweden Democrats 4 = Feminist Initiative 5 = Feminist Initiative and the Left Party 6 = The seven traditional parties 7 = Sweden Democrats and the Right.	***Only to be coded "yes" if it is obvious,** e.g. when the parties or their representatives are mentioned, either explicitly or through paraphrases*

	Aggressiveness (tone)		
	0 = Not at all aggressive 1 = Partly aggressive 2 = Predominantly aggressive.	*In what **tone** is the post as a whole written?* *- If **some part** of the post is read as aggressive, it should be coded **partly aggressive**.* *- If the post contains **mostly aggressive** text, it should be coded **predominantly** aggressive.*	
	HATRED HAT		
6	Another Flashback user	0 = No 1 = Yes	*If the post contains **words or statements** that indicate persecution (in the broad sense of the term) of **a group or an individual,** it should be coded "yes", Possible examples are:* *- threat* *- expressions of disrespect* *- insults* *- verbal violations* ***Use the coding "yes" also for isolated hateful statements. It doesn't have to be blatant.***
7	Specific public person		
8	Persons with specific sex/gender		
9	Persons who were born abroad, or whose parents are born abroad		
10	Persons with a specific ethnicity		
11	Persons with a specific sexual inclination		
12	Persons with specific skin color		
13	Something else		
14	If the hatred is pointed toward something else, please specify	Text	
	HOT		
15	Another Flashback user	0 = No 1 = Yes	*Does the post contain words that are explicit threats or assault? Assault means that someone threatens to **harm an individual or his or her property.** The assault **can be directed to other persons, animals or objects** that are important to the individual.*
16	Specific public person		
17	Persons with specific sex/gender		
18	Persons who were born abroad, or whose parents are born abroad		
19	Persons with a specific ethnicity		
20	Persons with a specific sexual inclination		
21	Persons with specific skin color		
22	Something else		
23	If the threat is pointed toward something else, please specify	Text	
24	Male preference		
	0 = No 1 = Yes	If the post contains **words that in any way state the superiority of men** over women, it should be coded "yes"	
25	Female preference		
	0 = No 1 = Yes	If the post contains **words that in any way state the superiority of women** over men, it should be coded "yes"	
26	Gender equality preference		
	0 = Nej 1 = Ja	If the post contains **words of men and women being equal**, it should be coded "yes"	
27	Attitudes towards foreigners		
	0 = No opinion 1 = Positive attitude 2 = Neutral attitude 3 = Negative attitude	By foreigners is meant people who are **born abroad or whose parents were born** abroad.	

28	Gender - disadvantaged	
	0 = No opinion 1 = Men are disadvantaged 2 = Women are disadvantaged	If the post contains words that express women as disadvantaged or men as disadvantaged, it shall be coded as 1 or 2 respectively.
29	Ethnicity - disadvantaged	
	0 = No opinion 1 = Swedes are disadvantaged 2 = Immigrants are disadvantaged	If the post contains words that express Swedes as disadvantaged or immigrants as disadvantaged it shall be coded as 1 or 2 respectively.
30	Us and Them	
	0 = No 1 = Yes	If the post explicitly contains a language of ”us and them” or clearly expresses an in-group out-group view the variable should be coded yes.
31	Sarcasm or irony	
	0 = No 1 = Yes partly 2 = Yes, fully	If the post contains sarcasm or irony in part or in full it should be coded yes.

References

Chen, Ying, Yilu Zhou, Sencun Zhu, and Heng Xu. 2012. "Detecting offensive language in social media to protect adolescent online safety." 71–80, IEEE.

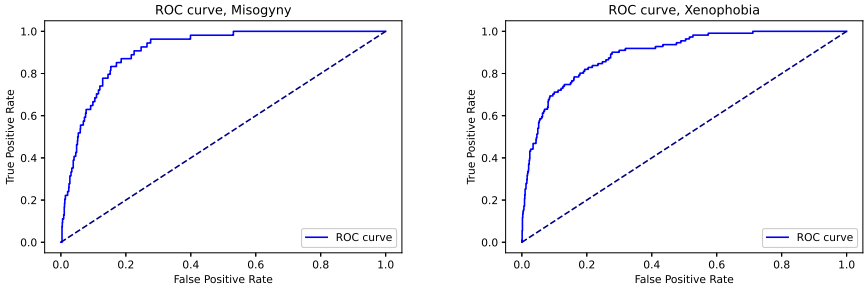
Cohen, Katie, Fredrik Johansson, Lisa Kaati, and Jonas Clausen Mork. 2014. "Detecting linguistic markers for radical violence in social media." *Terrorism and Political Violence*, 26(1): 246–256.

Do, Salomé, Étienne Ollion, and Rubing Shen. 2024. "The Augmented Social Scientist: Using Sequential Transfer Learning to Annotate Millions of Texts with Human-level Accuracy." *Sociological Methods & Research* 53(3):1167–1200.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2013. *An introduction to statistical learning.* Vol. 112, Springer.

S.8 Figures and tables

S.8.1 Figures



(a) (b)

Figure S.1: ROC Curves

Figure displaying the ROC-curves of the different prediction models. The curve display the set of combinations of true positive and false positive rates for the model for different classification cut offs. The higher the curve is above the 45-degree line, the better the predictions will be. The area under the ROC curve (AUC) for is 0.69 for misogynism and 0.75 for xenophobia.

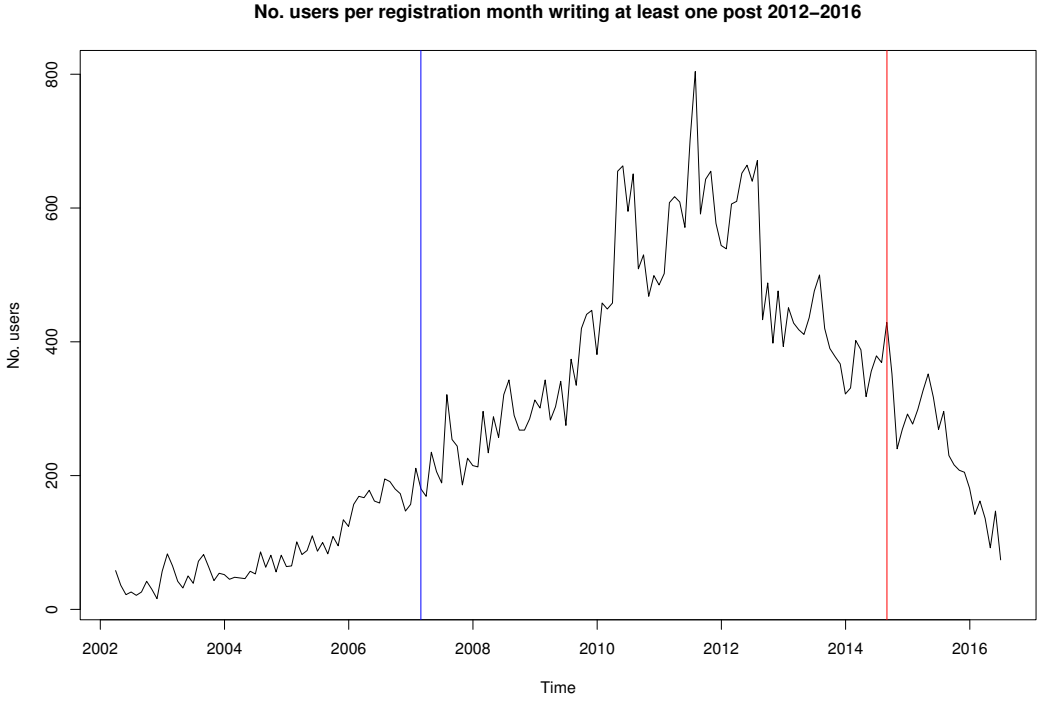
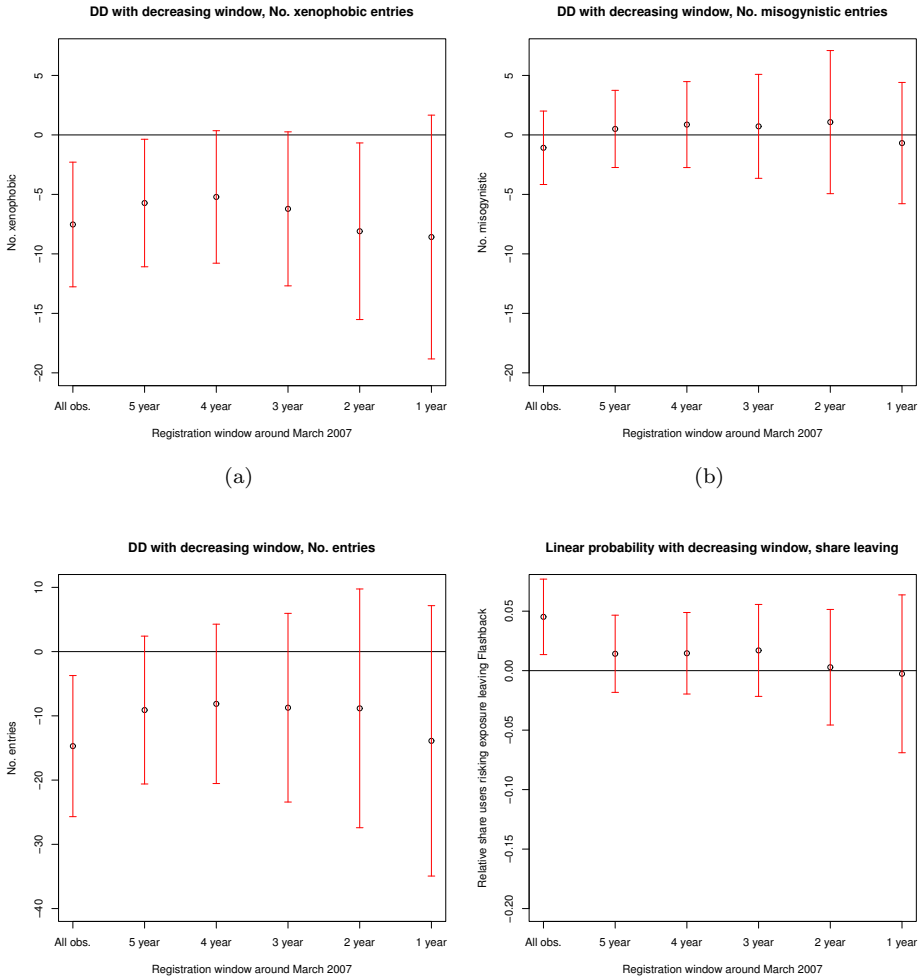


Figure S.2: Users per registration month in data

The figure presents the number of users per registration month who have written at least one post in our data from 2012 to 2016. The red line lies on September 2014 and thus represents where the period after the event starts. The blue line lies on March 2007. The users below this cut-off registration date are the ones that risked exposure after September 10 2014.

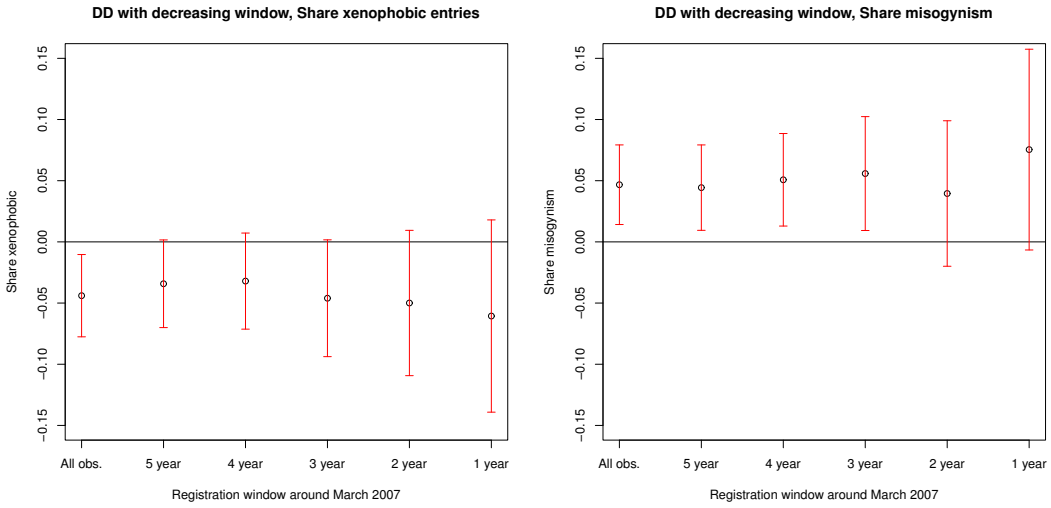


(a) (b)

(c) (d)

Figure S.3: Difference-in-difference estimates varying the distance to the date defining risk of exposure

The graphs present the coefficients and 95% confidence intervals from separate difference-in-difference regressions in which the distance to the registration date defining assignment into the group risking exposure, March 2007, is allowed to shrink. The first coefficient and confidence interval thus uses all available data, the second only includes users registered from March 2002 to March 2012, the third includes users registered from March 2003 to March 2011, the fourth includes users registered from March 2004 to March 2010, the fifth includes users registered from March 2005 to March 2009, and finally the sixth estimate includes users registered from March 2006 to March 2008. The period from which the posts are collected is from 2012-2016 in each regression, with standard errors clustered at the user level. All regressions include linear and quadratic controls in the user start date.



(a) (b)

Figure S.4: Difference-in-difference estimates varying the distance to the date defining risk of exposure, unbalanced panel

The panels present the coefficients and 95% confidence intervals from separate difference-in-difference regressions in which the distance to the registration date defining assignment into the group risking exposure, March 2007, is allowed to shrink. The first coefficient and confidence interval thus uses all available data, the second only includes users registered from March 2002 to March 2012, the third includes users registered from March 2003 to March 2011, the fourth includes users registered from March 2004 to March 2010, the fifth includes users registered from March 2005 to March 2009 and finally the sixth estimate includes users registered from March 2006 to March 2008. The time period from which the posts are collected is from 2012-2016 in each regression, with standard errors clustered at the user level. All regressions include linear and quadratic controls in user startdate.

S.8.2 Tables

Table S.1: Confusion Matrix Xenophobia

	Predicted not xenophobic	Predicted xenophobic
Actual not xenophobic	888	213
Actual xenophobic	20	91

Table S.2: Confusion Matrix Misogyny

	Predicted not misogynistic	Predicted misogynistic
Actual not misogynistic	838	320
Actual misogynistic	2	52

Table S.3: Comparing users staying to users leaving after the event

	Mean	SD
Panel A: Balanced panel prior to event date		
Share xenophobic	0.298	0.282
Share misogynistic	0.230	0.261
Registration date	2010.364	2.773
Share exposed users	0.148	0.355
Users	11159	
Panel B: Users that do not post after the event		
Share xenophobic	0.311	0.362
Share misogynistic	0.253	0.343
Registration date	2010.974	2.335
Share exposed users	0.083	0.276
Users	23301	

The table presents the summary statistics for the balanced panel (Panel A) compared to the drop outs (Panel B), which are excluded from the balanced panel. The xenophobic and misogynism variables are based on predictions from fine tuned BERT models.

Table S.4: Summary statistics pre and post event by group

	Risk exposure, pre	Risk exposure, post	Not at risk, pre	Not at risk, post
No. entries	70.978	53.207	59.355	56.295
No. xenophobic	24.310	16.862	18.365	18.447
No. Misogynistic	15.063	11.352	16.257	13.628
Share xenophobic	0.275	0.278	0.302	0.293
Share misogynistic	0.205	0.169	0.234	0.201

The table presents the summary statistics for the lasso predicted dummy variables of xenophobia and misogyny respectively.

Table S.5: Summary statistics pre and post event by group, classified data from the RA

	Risk exposure, pre	Risk exposure, post	Not at risk, pre	Not at risk, post
No. entries	1065.00	96.00	1975.00	904.00
No. users	500.00	54.00	1054.00	503.00
No. threads	180.00	37.00	181.00	84.00
Hate	0.21	0.14	0.23	0.28
Xenophobia	0.08	0.03	0.10	0.11
Misogynism	0.03	0.04	0.06	0.09
Threat	0.01	0.00	0.01	0.01
Aggresive	0.56	0.41	0.63	0.67

All variables in the table are dummy variables.
The share of misogynistic and xenophobic entries do not sum to the total share of hateful entries due to the fact that there is also general hateful comments and hate towards particular individuals, such as politicians and celebrities, in the data as well.

Table S.6: Impact of a decrease in anonymity using only RA data

	Dependent variable:				
	Share hate	Share Xenophobia	Share misogyny	Share threat	Share aggression
	(1)	(2)	(3)	(4)	(5)
Post*Treated	-0.128^{**}	-0.064^{*}	-0.024	-0.010	-0.187^{*}
	(0.043)	(0.025)	(0.024)	(0.005)	(0.094)
Post reveal	0.050^{*}	0.011	0.034^{**}	0.0002	0.038
	(0.020)	(0.015)	(0.012)	(0.004)	(0.039)
Treated	-0.015	-0.012	-0.024^{**}	0.001	-0.079^{*}
	(0.019)	(0.013)	(0.009)	(0.004)	(0.036)
Constant	0.228^{***}	0.096^{***}	0.055^{***}	0.009^{***}	0.635^{***}
	(0.011)	(0.008)	(0.006)	(0.002)	(0.022)
Observations	4040	4040	4040	4040	4040

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The table presents results from regressions utilizing the standard diff-in-diff model using only the data annotated by the RA. Standard errors clustered at the user level.

Table S.7: Differences in drop-out rate between those at risk and not at risk of exposure

	Dependent variable:		
	P(Leaving forum)		
	(1)	(2)	(3)
Risk exposure	-0.152^{***}	-0.056^{***}	0.045^{**}
	(0.009)	(0.012)	(0.016)
User startdate		0.016^{***}	16.818^{***}
		(0.001)	(1.727)
User startdate squared			-0.004^{***}
			(0.0004)
Constant	0.692^{***}	-32.484^{***}	$-16{,}923.990^{***}$
	(0.003)	(2.985)	(1,736.179)
Observations	34460	34460	34460

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The sample consists of all users writing at least one post before the event in our data. One obseration is a user. The dependent variable is the probability that a user drops out of the forum from the pre to the post event period.

Table S.8: Pre-event differences between those at risk and not at risk of exposure

	Dependent variable:				
	No. xenophobic	Share xenophobic	No. misogynistic	Share misogynistic	No. entries
	(1)	(2)	(3)	(4)	(5)
	Panel A: Basic difference in means				
Risk exposure	5.945	-0.027^{***}	-1.194	-0.029^{***}	11.623
	(3.264)	(0.007)	(2.064)	(0.007)	(7.290)
	Panel B:Linear control in startdate				
Risk exposure	-0.551	-0.005	-4.223	-0.016	-12.174
	(4.015)	(0.011)	(3.847)	(0.010)	(10.932)
User startdate	-1.135^{***}	0.004^{***}	-0.529	0.002^{*}	-4.159^{***}
	(0.379)	(0.001)	(0.528)	(0.001)	(1.282)
	Panel C: Quadratic control in startdate				
Risk exposure	3.260	0.023^{*}	-2.645	-0.011	-6.813
	(6.641)	(0.014)	(3.673)	(0.012)	(15.147)
User startdate	644.412	4.750^{***}	266.697	0.881	903.854
	(736.971)	(1.457)	(357.090)	(1.362)	(1,656.171)
User startdate squared	-0.161	-0.001^{***}	-0.066	-0.0002	-0.226
	(0.183)	(0.0004)	(0.089)	(0.0003)	(0.412)
Observations	11159	11159	11159	11159	11159

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until September 10th 2014. It thus covers the the period in our data before the event. Standard errors clustered at the user level. The dependent variable is the number of individual xenophobic entries prior to the event in column 1, the share of xenophobic entries in column 2, the number of misogynistic entries in columne 3, the share of misogynistic comments in column 4 and the number of entries in column 5.

Table S.9: Impact of a decrease in anonymity on xenophobic and misogynistic writing

	Dependent variable:				
	No. xenophobic	Share xenophobic	No. misogynistic	Share misogynistic	No. entries
	(1)	(2)	(3)	(4)	(5)
	Panel A: Basic difference-in-difference				
Post*Risk exposure	−7.529***	0.012	−1.082	−0.003	−14.711***
	(2.673)	(0.009)	(1.575)	(0.008)	(5.603)
Post reveal	0.082	−0.009**	−2.630***	−0.033****	−3.060
	(0.764)	(0.004)	(0.966)	(0.003)	(2.348)
Risk exposure	5.945*	−0.027****	−1.194	−0.029****	11.623
	(3.264)	(0.007)	(2.064)	(0.007)	(7.290)
	Panel B: Linear control in startdate				
Post*Risk exposure	−7.529***	0.012	−1.082	−0.003	−14.711***
	(2.673)	(0.009)	(1.575)	(0.008)	(5.603)
Post reveal	0.082	−0.009**	−2.630***	−0.033****	−3.060
	(0.764)	(0.004)	(0.966)	(0.003)	(2.348)
Risk exposure	3.485	−0.002	−0.441	−0.014	0.798
	(3.823)	(0.010)	(3.388)	(0.009)	(9.976)
User startdate	−0.430	0.004****	0.132	0.003***	−1.892*
	(0.329)	(0.001)	(0.457)	(0.001)	(1.086)
	Panel C: Quadratic control in startdate				
Post*Risk exposure	−7.529***	0.012	−1.082	−0.003	−14.711***
	(2.673)	(0.009)	(1.575)	(0.008)	(5.603)
Post reveal	0.082	−0.009**	−2.630***	−0.033****	−3.060
	(0.764)	(0.004)	(0.966)	(0.003)	(2.349)
Risk exposure	3.171	0.011	−2.263	−0.010	−5.237
	(5.244)	(0.012)	(3.087)	(0.011)	(12.465)
User startdate	−53.668	2.218*	−308.615	0.642	−1,023.948
	(524.785)	(1.218)	(326.559)	(1.042)	(1,289.583)
User startdate squared	0.013	−0.001*	0.077	−0.0002	0.254
	(0.130)	(0.0003)	(0.081)	(0.0003)	(0.321)
Observations	22318	22318	22318	22318	22318

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until September 10th 2014. It thus covers the the period in our data before the event. Standard errors clustered at the user level. The dependent variable is the number of individual xenophobic entries prior to the event in column 1, the share of xenophobic entries in column 2, the number of misogynistic entries in columne 3, the share of misogynistic comments in column 4 and the number of entries in column 5.

Table S.10: Impact of a decrease in anonymity on xenophobic and misogynistic writing, predictions from LASSO model

	Dependent variable:				
	No. xenophobic	Share xenophobic	No. misogynistic	Share misogynistic	No. entries
	(1)	(2)	(3)	(4)	(5)
	Panel A: Basic difference-in-difference				
Post*Risk exposure	−3.814**	−0.007	0.118	−0.0001	−14.711***
	(1.511)	(0.006)	(1.017)	(0.005)	(5.603)
Post reveal	−1.061***	−0.015****	−1.442**	−0.012****	−3.060
	(0.402)	(0.003)	(0.648)	(0.002)	(2.348)
Risk exposure	3.139*	−0.013***	−1.108	−0.021****	11.623
	(1.907)	(0.005)	(1.392)	(0.004)	(7.290)
	Panel B: Linear control in startdate				
Post*Risk exposure	−3.814**	−0.007	0.118	−0.0001	−14.711***
	(1.511)	(0.006)	(1.017)	(0.005)	(5.603)
Post reveal	−1.061***	−0.015****	−1.442**	−0.012****	−3.060
	(0.402)	(0.003)	(0.648)	(0.002)	(2.348)
Risk exposure	1.927	−0.007	0.016	−0.007	0.798
	(2.190)	(0.006)	(2.262)	(0.006)	(9.976)
User startdate	−0.212	0.001	0.196	0.002***	−1.892*
	(0.176)	(0.001)	(0.322)	(0.001)	(1.086)
	Panel C: Quadratic control in startdate				
Post*Risk exposure	−3.814**	−0.007	0.118	−0.0001	−14.711***
	(1.511)	(0.006)	(1.017)	(0.005)	(5.603)
Post reveal	−1.061***	−0.015****	−1.442**	−0.012****	−3.060
	(0.402)	(0.003)	(0.648)	(0.002)	(2.349)
Risk exposure	1.881	0.006	−1.467	−0.009	−5.237
	(3.141)	(0.008)	(2.039)	(0.007)	(12.460)
User startdate	−8.058	2.207***	−250.865	−0.384	−1,023.948
	(331.353)	(0.771)	(253.640)	(0.757)	(1,288.262)
User startdate squared	0.002	−0.001***	0.062	0.0001	0.254
	(0.082)	(0.0002)	(0.063)	(0.0002)	(0.320)
Observations	22318	22318	22318	22318	22318

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until September 10th 2014. It thus covers the the period in our data before the event. Standard errors clustered at the user level. The dependent variable is the number of individual xenophobic entries prior to the event in column 1, the share of xenophobic entries in column 2, the number of misogynistic entries in columne 3, the share of misogynistic comments in column 4 and the number of entries in column 5.

Table S.11: Impact of a decrease in anonymity on number of written entries, by level of hate prior to event

	Dependent variable:			
	No. entries			
	(1)	(2)	(3)	(4)
	Basic difference-in-difference			
Post*Risked exposure	−1.346	−28.662***	−13.133**	−17.714*
	(5.657)	(9.900)	(6.330)	(9.593)
Post reveal	−5.482	−0.793	1.742	−7.712**
	(3.637)	(3.014)	(2.531)	(3.919)
Risked exposure	−4.422	29.191**	15.249*	10.707
	(7.717)	(12.651)	(9.247)	(11.415)
	Linear control in startdate			
Post*Risked exposure	−1.346	−28.662***	−13.133**	−17.714*
	(5.658)	(9.901)	(6.330)	(9.594)
Post reveal	−5.482	−0.793	1.742	−7.712**
	(3.637)	(3.015)	(2.531)	(3.919)
Risked exposure	−7.144	9.434	−3.182	8.254
	(13.112)	(15.186)	(11.504)	(16.407)
User startdate	−0.472	−3.478**	−3.193***	−0.433
	(1.619)	(1.433)	(1.128)	(1.876)
	Quadratic control in startdate			
Post*Risked exposure	−1.346	−28.662***	−13.133**	−17.714*
	(5.658)	(9.901)	(6.330)	(9.594)
Post reveal	−5.482	−0.793	1.742	−7.712**
	(3.637)	(3.015)	(2.531)	(3.920)
Risked exposure	−12.173	0.004	−8.654	−0.438
	(12.915)	(21.771)	(17.729)	(17.189)
User startdate	−843.186	−1,619.941	−926.149	−1,483.162
	(1,183.184)	(2,410.508)	(2,022.457)	(1,502.682)
User startdate squared	0.210	0.402	0.230	0.369
	(0.294)	(0.599)	(0.503)	(0.374)
Sample	Xenophobia below median	Xenophobia above median	Misogynism below median	Misogynism above median
Observations	10900	11418	11158	11160

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The time period used in the regressions is from January 1st 2012 until December 31st 2016. Standard errors clustered at the user level. Each column uses the data from either above or below of the median in the distribution of individual share xenophobia and share misogyny in the pre-treatment period, respectively. The outcome variable is individual number of entries in the pre- and post-period

Table S.12: Impact of a decrease in anonymity on xenophobic and misogynistic writing, robustnes results for unbalanced panel

Robustness results for share xenophobia						
Post*Risked exposure	-0.054^{**}	-0.044^{**}	-0.044^{*}	-0.051^{**}	-0.038^{*}	-0.052^{**}
	(0.017)	(0.017)	(0.017)	(0.018)	(0.017)	(0.017)
Post reveal	0.028^{***}	0.019^{**}	0.018^{*}	0.044^{***}	0.012	0.032^{***}
	(0.007)	(0.007)	(0.007)	(0.008)	(0.007)	(0.007)
Risked exposure	0.028	0.068^{**}	0.064^{*}	0.028	0.028	0.028
	(0.018)	(0.024)	(0.031)	(0.018)	(0.018)	(0.018)
User startdate		0.007^{**}	−0.467			
		(0.002)	(1.968)			
User startdate squared			0.0001			
			(0.0005)			
Robustness results for share misogynism						
Post*Risked exposure	0.036^{*}	0.045^{**}	0.047^{**}	0.040^{*}	0.040^{*}	0.037^{*}
	(0.016)	(0.016)	(0.017)	(0.018)	(0.017)	(0.017)
Post reveal	-0.035^{***}	-0.044^{***}	-0.046^{***}	-0.027^{**}	-0.040^{***}	-0.033^{***}
	(0.008)	(0.008)	(0.009)	(0.009)	(0.010)	(0.009)
Risked exposure	-0.069^{***}	−0.032	−0.047	-0.069^{***}	-0.069^{***}	-0.069^{***}
	(0.015)	(0.023)	(0.030)	(0.015)	(0.015)	(0.015)
User startdate		0.007^{*}	−1.818			
		(0.003)	(2.146)			
User startdate squared			0.0005			
			(0.001)			
Drop period between events	No	No	No	Yes	No	No
Last registered	-	-	-	-	Aug 2014	-
Exclude election week	No	No	No	No	No	Yes
Observations	1984224	1984224	1984224	1222891	1754758	1961610

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
The standard errors are clustered at the user level. The dependent variable is a dummy indicating if a post is predicted as xenophobic (Panel A) or misogynistic (Panel B).

Table S.13: Summary statistics pre and post event by group, unbalanced panel

	Risk exposure, pre	Risk exposure, post	Not at risk, pre	Not at risk, post
No. entries	145385	98219	902416	838204
No. users	3588	2536	30872	22835
No. threads	13089	9904	18111	15563
Xenophobia	0.34	0.32	0.32	0.34
Misogynism	0.21	0.21	0.28	0.25

The table presents the summary statistics for the lasso predicted dummy variables of xenophobia and misogyny respectively.

Table S.14: Simple correlations

	Dependent variable:					
	Individual share xenophobia	Individual share misogyny	Misogynistic entry	Individual share misogynism	Xenophobic entry	Misogynistic entry
	(1)	(2)	(3)	(4)	(5)	(6)
No. individual entries	0.00001*					
	(0.00001)					
No. individual entries		0.00002***				
		(0.00001)				
Xenophobic entry			0.030***			
			(0.001)			
Individual share xenophobia				0.041***		
				(0.004)		
First entry is xenophobic					0.258***	
					(0.001)	
First entry is misogynistic						0.389***
						(0.001)
Constant	0.312***	0.232***	0.248***	0.220***	0.199***	0.136***
	(0.002)	(0.001)	(0.0004)	(0.002)	(0.0005)	(0.0004)
Observations	48672	48672	1984224	48672	1748688	1748688

Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Standard errors clustered at the thread level in the 7th and 8th column.